

US Patent & Trademark Office

Patent Public Search | Text View

United States Patent Application Publication

20250285888

Kind Code

A1

Publication Date

September 11, 2025

Inventor(s)

SUZUKI; Yoshikuni

LASER PROCESSING APPARATUS, LASER PROCESSING METHOD, LASER PROCESSING PROGRAM, RECORDING MEDIUM, SEMICONDUCTOR CHIP MANUFACTURING METHOD AND SEMICONDUCTOR CHIP

Abstract

The length (constant velocity distance) of the constant velocity zone is set according to the length of the planned dividing line in the X direction, and the processing velocity in the irradiation position scan is adjusted according to the length (constant velocity distance) of the constant velocity zone (velocity adjustment processing). Therefore, it is possible to suppress a total time (scanning time) of the acceleration period, the constant velocity period and the deceleration period. In this way, the planned dividing line can be efficiently processed in the laser processing technique for processing the planned dividing line by moving the laser beam along the planned dividing line of the semiconductor substrate.

Inventors: SUZUKI; Yoshikuni (Iwata-shi, Shizuoka, JP)

Applicant: YAMAHA HATSUDOKI KABUSHIKI KAISHA (Iwata-shi, Shizuoka-ken, JP)

Family ID: 1000008626156

Assignee: YAMAHA HATSUDOKI KABUSHIKI KAISHA (Iwata-shi, Shizuoka-ken, JP)

Appl. No.: 18/857213

Filed (or PCT Filed): April 18, 2022

PCT No.: PCT/JP2022/018063

Publication Classification

Int. Cl.: H01L21/67 (20060101); **B23K26/08** (20140101); **H01L21/78** (20060101)

U.S. Cl.:

CPC H01L21/67092 (20130101); **B23K26/0876** (20130101); **H01L21/78** (20130101);

Background/Summary

CROSS-REFERENCE TO RELATED APPLICATION [0001] This application is a National Stage of International Patent Application No. PCT/JP2022/018063, filed Apr. 18, 2022, the entire contents of which is incorporated herein by reference.

BACKGROUND

Technical Field

[0002] This disclosure relates to a technique for processing a processing line by irradiating a laser beam to the processing line provided on a processing object.

Background Art

[0003] A Publication of Japanese Patent No. 5804716, a Publication of Japanese Patent No. 5554593 and a Publication of Japanese Patent No. 5037082 describe a laser processing technique for processing a planned dividing line by relatively moving a laser beam with respect to a semiconductor substrate while irradiating the laser beam to the planned dividing line provided on the semiconductor substrate. For example, as shown in the Publication of Japanese Patent No. 5804716, a plurality of planned dividing lines are processed in turn by reciprocating a laser beam while changing the planned dividing line, to which the laser beam is irradiated, on forward and return paths in this laser processing technique.

SUMMARY

[0004] In the laser processing technique as described above, a processing line is processed by irradiating a laser beam to a laser irradiation position while relatively moving the laser irradiation position of a processing head for irradiating the laser beam to the predetermined laser irradiation position with respect to a processing object (semiconductor substrate). That is, the laser beam is irradiated to the laser irradiation position moving along the processing line (planned dividing line) of the processing object while being moved from one side toward the other side of the processing object. At this time, it is required to move the laser beam at a constant processing velocity with respect to the processing line. Thus, a velocity of the laser irradiation position is accelerated from zero to a processing velocity by the time the laser irradiation position, which started to move toward the processing object from the one side of the processing object, reaches the processing object. Further, the velocity of the laser irradiation position, which has passed through the processing object toward the other side, is decelerated from the processing velocity to zero.

[0005] A high processing velocity is advantageous to complete a processing of moving the laser beam along the processing line in a short time. However, if the processing velocity is high, a time required to accelerate the laser irradiation position from zero to the processing velocity and a time required to decelerate the laser irradiation position from the processing velocity to zero take long. Thus, it has taken time to process the processing line in some cases since the processing velocity is improper.

[0006] This disclosure was developed in view of the above problem and aims to provide a technique for enabling a processing line to be efficiently processed in a laser processing technique for processing the processing line by moving a laser beam along the processing line of a processing object.

[0007] A laser processing apparatus according to the disclosure comprises a supporting member

supporting a processing object having a plurality of processing lines parallel to each other such that the processing lines are parallel to a predetermined processing direction; a processing head including a laser light source emitting a laser beam, the processing head irradiating the laser beam to a laser irradiation position from the laser light source; and a processing-axis driver relatively moving the laser irradiation position in the processing direction with respect to the processing object by driving at least one of the supporting member and the processing head in the processing direction. The laser processing apparatus also comprises a control unit for performing an irradiation position scan to process one target line, out of the plurality of processing lines, by irradiating the laser beam from the processing head to the laser irradiation position moving along the one target line while causing the processing-axis driver to move the laser irradiation position with respect to the processing object from a start point on one side of the processing object to an end point on the other side opposite to the one side of the processing object in the processing direction. In the irradiation position scan, a constant velocity zone including the target line being set between the start point and the end point in the processing direction, a velocity of the laser irradiation position in the processing direction with respect to the processing object increasing from zero to a processing velocity in a first period of moving the laser irradiation position from the start point to an end of the constant velocity zone on the one side, the laser irradiation position moving at the constant processing velocity in the processing direction with respect to the processing object through a second period of moving the laser irradiation position from the end on the one side to an end on the other side of the constant velocity zone, and the velocity of the laser irradiation position in the processing direction with respect to the processing object decreasing from the processing velocity to the zero in a third period from arrival of the laser irradiation position at the end of the constant velocity zone on the other side to arrival of the laser irradiation position at the end point. Also, the control unit sets a length of the constant velocity zone according to a length of the target line in the processing direction and performs a velocity adjustment processing of adjusting the processing velocity in the irradiation position scan according to the length of the constant velocity zone.

[0008] A laser processing method according to the disclosure comprises supporting a processing object having a plurality of processing lines parallel to each other by a supporting member such that the processing lines are parallel to a predetermined processing direction; and performing an irradiation position scan to process one target line, out of the plurality of processing lines, by irradiating a laser beam to a laser irradiation position from a processing head including a laser light source for emitting the laser beam. The processing head irradiates the laser beam from the laser light source to the laser irradiation position, while moving the laser irradiation position with respect to the processing object from a start point on one side of the processing object to an end point on the other side, being opposite to the one side, of the processing object, in the processing direction by a processing-axis driver for relatively moving the laser irradiation position in the processing direction with respect to the processing object by driving at least one of the processing head and the supporting member in the processing direction. In the irradiation position scan, a constant velocity zone including the target line being set between the start point and the end point in the processing direction, a velocity of the laser irradiation position in the processing direction with respect to the processing object increasing from zero to a processing velocity in a first period of moving the laser irradiation position from the start point to an end of the constant velocity zone on the one side, the laser irradiation position moving at the constant processing velocity in the processing direction with respect to the processing object through a second period of moving the laser irradiation position from the end on the one side to an end on the other side of the constant velocity zone, and the velocity of the laser irradiation position in the processing direction with respect to the processing object decreasing from the processing velocity to the zero in a third period from arrival of the laser irradiation position at the end of the constant velocity zone on the other side to arrival of the laser irradiation position at the end point. A length of the constant velocity zone is set

according to a length of the target line in the processing direction, and the processing velocity in the irradiation position scan is adjusted according to the length of the constant velocity zone.

[0009] In the disclosure (laser processing apparatus and laser processing method) thus configured, the one target line, out of the plurality of processing lines, is processed by irradiating the laser beam from the processing head to the laser irradiation position moving along the one target line while moving the laser irradiation position with respect to the processing object from the start point on the one side to the end point on the other side of the processing object in the processing direction (irradiation position scan). In this irradiation position scan, the constant velocity zone including the target line is set between the start point and the end point in the processing direction. The velocity of the laser irradiation position in the processing direction with respect to the processing object increases from zero to the processing velocity in the first period of moving the laser irradiation position from the start point to the end of the constant velocity zone on the one side. Further, the laser irradiation position moves at the constant processing velocity in the processing direction with respect to the processing object through the second period of moving the laser irradiation position from the end on the one side to the end on the other side of the constant velocity zone. Furthermore, the velocity of the laser irradiation position in the processing direction with respect to the processing object decreases from the processing velocity to zero in the third period from the arrival of the laser irradiation position at the end of the constant velocity zone on the other side to the arrival thereof at the end point. At this time, the length of the constant velocity zone is set according to the length of the target line in the processing direction, and the processing velocity in the irradiation position scan is adjusted according to the length of the constant velocity zone (velocity adjustment processing). Therefore, a total period of the first period, the second period and the third period can be suppressed. In this way, the processing line can be efficiently processed in a laser processing technique for processing the processing line by moving the laser beam along the processing line of the processing object.

[0010] The laser processing apparatus may be configured so that the control unit performs a plurality of the irradiation position scans by repeating the irradiation position scan while changing the target line among the plurality of processing lines and performs the velocity adjustment processing for each of the plurality of irradiation position scans. In such a configuration, each of the plurality of processing lines can be efficiently processed and the processing of the processing object can be quickly completed.

[0011] The laser processing apparatus may be configured so that the control unit performs a frequency adjustment processing of adjusting a frequency of the laser beam to be irradiated to the laser irradiation position moving along the target line in the irradiation position scan according to the processing velocity in the irradiation position scan. In such a configuration, the processing line can be precisely processed by irradiating the laser beam of a proper frequency corresponding to the adjusted processing velocity to the processing line.

[0012] The laser processing apparatus may be configured so that the control unit performs a plurality of the irradiation position scans by repeating the irradiation position scan while changing the target line among the plurality of processing lines and performs the frequency adjustment processing for each of the plurality of irradiation position scans. In such a configuration, each of the plurality of processing lines can be precisely processed.

[0013] The laser processing apparatus may be configured so that the control unit sets a common processing velocity, the common processing velocity being a processing velocity common to a plurality of the irradiation position scans performed by repeating the irradiation position scan while changing the target line among the plurality of processing lines, and performs each of the plurality of irradiation position scans with the processing velocity set to the common processing velocity. Also, the common processing velocity is adjusted according to the length of the constant velocity zone in each of the plurality of irradiation position scans in the velocity adjustment processing. In such a configuration, the plurality of processing lines can be efficiently processed without

switching the processing velocity among the plurality of processing lines, and the processing of the processing object can be quickly completed.

[0014] The laser processing apparatus may be configured so that the control unit obtains a common frequency common to the plurality of irradiation position scans according to the common processing velocity and causes the laser beam of the common frequency to be irradiated to the laser irradiation position in each of the plurality of irradiation position scans. In such a configuration, the frequency of the laser beam needs not be switched among the plurality of processing lines, and it is possible to eliminate an influence of a time required to switch the frequency on the completion of the processing for the processing object.

[0015] The laser processing apparatus may further comprises a storage for storing correspondence relationship information representing a correspondence relationship of a size, the common processing velocity and the common frequency for each of a plurality of sizes of the processing objects. The control unit adjusts the common processing velocity and the common frequency based on the size of the processing object as a target of the irradiation position scan and the correspondence relationship information. In such a configuration, the common processing velocity and the common frequency can be easily adjusted by referring to the correspondence relationship information.

[0016] Specifically, the laser processing apparatus may be configured such that the correspondence relationship information represents a diameter of the processing object as the size and represents a correspondence relationship of the size, the common processing velocity and the common frequency for each of a diameter of 200 mm and a diameter of 300 mm.

[0017] Further, various setting modes of the constant velocity zone for the target line are supposed. In short, the constant velocity zone only has to include the target line in the processing direction. Therefore, both ends of the constant velocity zone may coincide with both ends of the target line in the processing direction.

[0018] A laser processing program according to the disclosure causes a computer to carry out the laser processing method described above.

[0019] A recording medium according to the disclosure computer-readably stores the laser processing program described above.

[0020] A semiconductor chip manufacturing method according to the disclosure, comprises processing a semiconductor substrate, having a plurality of semiconductor chips demarcated by processing lines and arrayed, by the laser processing method described above; and separating each of the plurality of semiconductor chips by expanding a tape, holding the semiconductor substrate by an adhesive force, processed by the laser processing method.

[0021] A semiconductor chip, according to the disclosure is manufactured by: processing a semiconductor substrate, having a plurality of semiconductor chips demarcated by processing lines and arrayed, by the laser processing method described above; and separating each of the plurality of semiconductor chips by expanding a tape, holding the semiconductor substrate by an adhesive force, processed by the laser processing method.

[0022] According to the disclosure, a processing line can be efficiently processed in a laser processing technique for processing the processing line by moving a laser beam along the processing line of a processing object.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0023] FIG. 1 is a front view schematically showing an example of a laser processing apparatus according to the disclosure;

[0024] FIG. 2 is a plan view schematically showing the laser processing apparatus of FIG.;

[0025] FIG. 3 is a block diagram showing the electrical configuration of the laser processing apparatus of FIG. 1;

[0026] FIG. 4 is a flow chart showing an example of a method for producing a laser processed substrate, for which the laser processing has been already performed;

[0027] FIG. 5 is a flow chart showing an example of the take-out of the ring frame;

[0028] FIG. 6 is a flow chart showing an example of the transfer of the ring frame;

[0029] FIG. 7A is a plan view schematically showing an example of an operation performed in accordance with the flow charts of FIGS. 5 and 6;

[0030] FIG. 7B is a plan view schematically showing an example of an operation performed in accordance with the flow charts of FIGS. 5 and 6;

[0031] FIG. 7C is a plan view schematically showing an example of an operation performed in accordance with the flow charts of FIGS. 5 and 6;

[0032] FIG. 7D is a plan view schematically showing an example of an operation performed in accordance with the flow charts of FIGS. 5 and 6;

[0033] FIG. 7E is a plan view schematically showing an example of an operation performed in accordance with the flow charts of FIGS. 5 and 6;

[0034] FIG. 8 is a flow chart showing an example of the storage of the ring frame;

[0035] FIG. 9 is a flow chart showing an example of the ring frame alignment;

[0036] FIG. 10 is a plan views schematically showing an example of an operation performed in the ring frame alignment;

[0037] FIG. 11 is a flow chart showing an example of the substrate processing;

[0038] FIG. 12 is a plan view schematically showing an example of an operation performed in accordance with the flow chart of FIG. 11;

[0039] FIG. 13A is a flow chart showing an example of the calibration;

[0040] FIG. 13B is a flow chart showing an example of stage plane specification performed in the calibration of FIG. 13A;

[0041] FIG. 13C is a flow chart showing an example of substrate plane specification performed in the calibration of FIG. 13A;

[0042] FIG. 14 is a flow chart showing a basic process of the line processing for each planned dividing line;

[0043] FIG. 15A is a chart schematically showing a first example of an operation performed in accordance with the flow chart of FIG. 14;

[0044] FIG. 15B is a chart schematically showing a second example of the operation performed in accordance with the flow chart of FIG. 14;

[0045] FIG. 15C is a chart schematically showing a third example of the operation performed in accordance with the flow chart of FIG. 14;

[0046] FIG. 15D is a chart schematically showing a fourth example of the operation performed in accordance with the flow chart of FIG. 14;

[0047] FIG. 15E is a chart schematically showing a fifth example of the operation performed in accordance with the flow chart of FIG. 14;

[0048] FIG. 15F is a chart schematically showing a sixth example of the operation performed in accordance with the flow chart of FIG. 14;

[0049] FIG. 15G is a chart schematically showing a seventh example of the operation performed in accordance with the flow chart of FIG. 14;

[0050] FIG. 16 is a flow chart showing a first application example of the line processing for each planned dividing line;

[0051] FIG. 17 is a chart schematically showing an example of an operation performed in accordance with the flow chart of FIG. 16;

[0052] FIG. 18 is a flow chart showing a second application example of the line processing for each planned dividing line;

[0053] FIG. **19A** is a chart schematically showing a first example of an operation performed in accordance with the flow chart of FIG. **18**;

[0054] FIG. **19B** is a chart schematically showing a second example of the operation performed in accordance with the flow chart of FIG. **18**;

[0055] FIG. **20** is a diagram schematically showing an example of an image of the semiconductor substrate obtained in Step **S1008** of FIG. **16** or Step **S1104** of FIG. **18**;

[0056] FIG. **21** is a flow chart showing an example of a determination method of laser processing conditions in the line processing;

[0057] FIG. **22A** is a chart showing parameters relating to the determination of laser processing conditions;

[0058] FIG. **22B** is a graph showing a time impact of the laser processing condition;

[0059] FIG. **22C** is a Table showing an example of a table to be referred to in the determination of the laser processing conditions of FIG. **21**;

[0060] FIG. **23** is a flow chart showing an example of the laser processing condition determination method for each semiconductor substrate;

[0061] FIG. **24** is a diagram showing an example of a table to be referred to in the laser processing condition determination of FIG. **23**;

[0062] FIG. **25** is a flow chart showing an example of laser processing condition setting to set the processing conditions obtained in advance by the laser processing condition method of FIG. **23** according to the diameter of the substrate; and

[0063] FIG. **26** is a diagram schematically showing a correspondence relationship table used in the flow chart of FIG. **25**.

DETAILED DESCRIPTION

[0064] FIG. **1** is a front view schematically showing an example of a laser processing apparatus according to the disclosure, and FIG. **2** is a plan view schematically showing the laser processing apparatus of FIG. **1**. In both figures and subsequent figures, an X direction, which is a horizontal direction, a Y direction, which is a horizontal direction orthogonal to the X direction, and a Z direction, which is a vertical direction, are shown as appropriate. Further, a (+X) side in the X direction (right side in FIG. **2**) and a (−X) side (left side in FIG. **2**) opposite to the (+X) side in the X direction are shown as appropriate, and a (+Y) side in the Y direction (upper side in FIG. **2**) and a (−Y) side (lower side in FIG. **2**) opposite to the (+Y) side in the Y direction are shown as appropriate.

[0065] The laser processing apparatus **1** processes a semiconductor substrate W by irradiating a laser beam to the semiconductor substrate W (processing object). This semiconductor substrate W is held by a ring frame Fr via a tape E. The tape E is a dicing tape or a bonding tape, and the front surface (upper surface) of the tape E is adhesive. The ring frame Fr has an outer shape obtained by cutting parts of a regular octagon shape to provide slits Fs, and a circular opening Fo is provided in a center of the ring frame Fr. The front surface of the tape E is facing the ring frame Fr from below to overlap the entire opening Fo, and the peripheral edge of the front surface of the tape E is bonded to the bottom surface of the ring frame Fr by an adhesive force. Further, the semiconductor substrate W is bonded to the front surface of the tape E by an adhesive force. The semiconductor substrate W is conveyed in the laser processing apparatus **1** while being held by the ring frame Fr via the tape E in this way. Note that the semiconductor substrate W has a front surface and a back surface opposite to the front surface, and an electronic circuit is formed on the front surface of the semiconductor substrate W, whereas the back surface of the semiconductor substrate W is flat. The downward facing front surface of the semiconductor substrate W is bonded to the front surface of the tape E. That is, the semiconductor substrate W is held with the back surface of the semiconductor substrate W facing upward.

[0066] The laser processing apparatus **1** is provided with a substrate storage part **2** for storing the semiconductor substrate W and a chuck stage **3** (supporting member) for holding the

semiconductor substrate W taken out from the substrate storage part 2. The laser processing apparatus 1 is provided with a base plate 11 having a flat plate shape, and the substrate storage part 2 and the chuck stage 3 are supported by the base plate 11. The chuck stage 3 is arranged on the (+X) side of the substrate storage part 2 in the X direction, and arranged on the (−Y) side of the substrate storage part 2 in the Y direction. A space on the (−X) side of the chuck stage 3 in the X direction and on the (−Y) side of the substrate storage part 2 in the Y direction is a substrate transfer region Aw.

[0067] The substrate storage part 2 includes a substrate storage cassette 21. The substrate storage cassette 21 includes a pair of side walls 22 provided on both sides in the X direction and an opening 23 provided between the side walls 22, and the opening 23 is facing toward the (−Y) side (i.e. toward the substrate transfer region Aw). The pair of side walls 22 are flat plates provided perpendicular to the X direction and facing each other in the X direction. Further, supporting projections 24 are provided inside each of the pair of side walls 22. A pair of the supporting projections 24 facing each other in the X direction are provided at the same height. The ring frame Fr holding the semiconductor substrate W can be inserted to a position above the pair of supporting projections 24 from the (−Y) side via the opening 23. Both ends in the X direction of the ring frame Fr inserted in this way are supported from below by the pair of supporting projections 24. That is, a side above the pair of supporting projections 24 functions as a slot 25 for storing the ring frame Fr, and the ring frame Fr inserted into the slot 25 from the (−Y) side via the opening 23 is supported by the pair of supporting projections 24 corresponding to this slot 25. Therefore, the semiconductor substrate W supported on the ring frame Fr can be stored into the substrate storage cassette 21 by inserting the ring frame Fr into the slot 25 of the substrate storage cassette 21, and the semiconductor substrate W can be taken out from the substrate storage cassette 21 by withdrawing the ring frame Fr from the slot 25 of the substrate storage cassette 21.

[0068] Further, the substrate storage cassette 21 includes a Z-axis slider 26 for supporting the substrate storage cassette 21 and a Z-axis driving mechanism 27 for driving the Z-axis slider 26 in the Z direction. The Z-axis driving mechanism 27 is a single-axis robot mounted on the base plate 11 and includes a Z-axis drive transmitter 271 for supporting the Z-axis slider 26 movably in the Z direction and a Z-axis cassette motor 272 for driving the Z-axis slider 26 supported on the Z-axis drive transmitter 271 in the Z direction. The Z-axis drive transmitter 271 includes a ball screw to be driven by the Z-axis cassette motor 272, and the Z-axis slider 26 is attached to a nut of the ball screw. However, a specific configuration of the Z-axis driving mechanism 27 is not limited to this example and may be a linear motor. Such a Z-axis driving mechanism 27 moves the substrate storage cassette 21 supported on the Z-axis slider 26 in the Z direction by driving the Z-axis slider 26 supported on the Z-axis drive transmitter 271 by the Z-axis cassette motor 272.

[0069] A substrate insertion height 211 is set for the substrate storage cassette 21, and the semiconductor substrate W can be inserted into and withdrawn from the slot 25 located at the substrate insertion height 211. Therefore, the slot 25, into which and from which the semiconductor substrate W is inserted and withdrawn, can be changed by moving the substrate storage cassette 21 in the Z direction by the Z-axis driving mechanism 27 to change the slot 25 located at the substrate insertion height 211, out of a plurality of the slots 25.

[0070] In contrast, the laser processing apparatus 1 is provided with a Y-axis conveying mechanism 4 for conveying the ring frame Fr in the Y direction between the slot 25 at the substrate insertion height 211 and the substrate transfer region Aw. The Y-axis conveying mechanism 4 includes a lift hand 41, a Y-axis slider 43 for supporting the lift hand 41 and a Y-axis driving mechanism 45 for driving the Y-axis slider 43 in the Y direction. The Y-axis driving mechanism 45 is a single-axis robot mounted on the base plate 11 by an unillustrated frame and includes a Y-axis drive transmitter 451 for supporting the Y-axis slider 43 movably in the Y direction and a Y-axis lift hand motor 452 for driving the Y-axis slider 43 supported on the Y-axis drive transmitter 451 in the Y direction. The Y-axis drive transmitter 451 includes a ball screw to be driven by the Y-axis lift hand motor 452,

and the Y-axis slider **43** is attached to a nut of the ball screw. However, a specific configuration of the Y-axis driving mechanism **45** is not limited to this example and may be a linear motor. Such a Y-axis driving mechanism **45** moves the lift hand **41** supported on the Y-axis slider **43** in the Y direction by driving the Y-axis slider **43** supported on the Y-axis drive transmitter **451** by the Y-axis lift hand motor **452**.

[0071] The lift hand **41** includes a base part **411** supported on the Y-axis slider **43** and a fork **412** projecting toward the (+Y) side from the base part **411**. The fork **412** is located at the substrate insertion height **211** and can hold the ring frame Fr from below. The Y-axis conveying mechanism **4** moves the ring frame Fr held by the fork **412** of the lift hand **41** between the substrate storage cassette **21** and the substrate transfer region Aw by driving the lift hand **41** in the Y direction by the Y-axis driving mechanism **45** as described later.

[0072] Further, the laser processing apparatus **1** is provided with an XZ-axis conveying mechanism **5** for conveying the ring frame Fr in the X direction between the lift hand **41** located in the substrate transfer region Aw and the chuck stage **3**. The XZ-axis conveying mechanism **5** includes a suction hand **51**, an X-axis slider **53** for supporting the suction hand **51** and an X-axis driver **55** for driving the X-axis slider **53** in the X direction. The X-axis driver **55** is a single-axis robot mounted on the base plate **11** by an unillustrated frame and includes an X-axis drive transmitter **551** for supporting the X-axis slider **53** movably in the X direction and an X-axis suction hand motor **552** for driving the X-axis slider **53** supported on the X-axis drive transmitter **551** in the X direction. The X-axis drive transmitter **551** includes a ball screw to be driven by the X-axis suction hand motor **552**, and the X-axis slider **53** is attached to a nut of the ball screw. However, a specific configuration of the X-axis driver **55** is not limited to this example and may be a linear motor. Such an X-axis driver **55** moves the suction hand **51** supported on the X-axis slider **53** in the X direction by driving the X-axis slider **53** supported on the X-axis drive transmitter **551** by the X-axis suction hand motor **552**.

[0073] Further, the XZ-axis conveying mechanism **5** includes a Z-axis slider **56** attached to the suction hand **51** and a Z-axis driver **58** for driving the Z-axis slider **56** in the Z direction with respect to the X-axis slider **53**. That is, the suction hand **51** is supported on the X-axis slider **53** via the Z-axis slider **56** and the Z-axis driver **58**. The Z-axis driver **58** is a single-axis robot mounted on the X-axis slider **53** and includes a Z-axis drive transmitter **581** for supporting the Z-axis slider **56** movably in the Z direction and a Z-axis suction hand motor **582** for driving the Z-axis slider **56** supported on the Z-axis drive transmitter **581** in the Z direction. The Z-axis drive transmitter **581** includes a ball screw to be driven by the Z-axis suction hand motor **582**, and the Z-axis slider **56** is attached to a nut of the ball screw. However, a specific configuration of the Z-axis driver **58** is not limited to this example and may be a linear motor. The Z-axis slider **56** extends to a side below the X-axis drive transmitter **551** from the Z-axis driver **58** and the suction hand **51** is attached to the lower end of the Z-axis slider **56**. Such a Z-axis driver **58** moves the suction hand **51** supported on the Z-axis slider **56** in the Z direction by driving the Z-axis slider **56** supported on the Z-axis drive transmitter **581** by the Z-axis suction hand motor **582**.

[0074] The suction hand **51** includes a base part **511** supported on the Z-axis slider **56** and an annular suction member **512** projecting toward the (+Y) side from the base part **511**. The annular suction member **512** has a circular annular shape, and a plurality of suction holes are open in a bottom surface **513** of the annular suction member **512**. The ring frame Fr can be held from above by the suction hand **51** by sucking the ring frame Fr by a negative pressure generated in each suction hole of the bottom surface **513** while bringing the bottom surface **513** of this annular suction member **512** into contact with the ring frame Fr from above. The XZ-axis conveying mechanism **5** moves the ring frame Fr held by the annular suction member **512** of the suction hand **51** between the substrate transfer region Aw and the chuck stage **3** by driving the suction hand **51** in the X direction by the X-axis driver **55** and driving the suction hand **51** in the Z direction by the Z-axis driver **58** as described later.

[0075] The chuck stage **3** includes a suction plate **31**, on which the ring frame Fr supporting the semiconductor substrate W via the tape E is placed. The suction plate **31** has a circular shape, and a plurality of suction holes are open in an upper surface **311** of the suction plate **31**. The tape E can be fixed to the suction plate **31** by sucking the tape E in contact with the upper surface **311** by a negative pressure generated in each suction hole of the upper surface **311** of the suction plate **31**. Further, the chuck stage **3** includes a plurality of clampers **32** provided on the peripheral edge of the suction plate **31**. This chuck stage **3** fixes the ring frame Fr to the suction plate **31** by causing the clampers **32** to face the ring frame Fr placed on the suction plate **31** from above and sandwiching the ring frame Fr between the clampers **32** and the suction plate **31**. Further, the chuck stage **3** releases the fixing of the ring frame Fr to the suction plate **31** by laterally retracting the clampers **32** from the ring frame Fr.

[0076] As just described, the chuck stage **3** holds the semiconductor substrate W supported on the ring frame Fr via the tape E by sucking the tape E by the suction plate **31** and fixing the ring frame Fr by the clampers **32**. By using the clampers **32** in combination in this way, the tape E can be sucked to the suction plate **31** with a weak suction force and an influence of the suction of the tape E on the semiconductor substrate W can be mitigated as compared to the case where the semiconductor substrate W is held only by the suction of the tape E to the suction plate **31**.

[0077] Further, the laser processing apparatus **1** is provided with an XY θ drive table **6** for supporting the chuck stage **3**. The XY θ drive table **6** is arranged on the base plate **11** and drives the chuck stage **3** in the X direction, the Y direction and a θ direction with respect to the base plate **11**. Here, the θ direction is a rotation direction about an axis of rotation parallel to the Z direction. That is, the XY θ drive table **6** includes a Y-axis guide **61** mounted on the base plate **11** in parallel to the Y direction, a Y-axis slider **62** supported movably in the Y direction by the Y-axis guide **61** and a Y-axis driver **63** for driving the Y-axis slider **62** in the Y direction. The Y-axis driver **63** is a single-axis robot mounted on the base plate **11** and includes a Y-axis drive transmitter **631** for supporting the Y-axis slider **62** movably in the Y direction and a Y-axis table motor **632** for driving the Y-axis slider **62** supported on the Y-axis drive transmitter **631** in the Y direction. The Y-axis drive transmitter **631** includes a ball screw to be driven by the Y-axis table motor **632**, and the Y-axis slider **62** is attached to a nut of the ball screw. However, a specific configuration of the Y-axis driver **63** is not limited to this example and may be a linear motor.

[0078] Further, the XY θ drive table **6** includes an X-axis slider **64** and an X-axis driver **65** for driving the X-axis slider **64** in the X direction with respect to the Y-axis slider **62**. The X-axis driver **65** is a single-axis robot mounted on the Y-axis slider **62** and includes an X-axis drive transmitter **651** for supporting the X-axis slider **64** movably in the X direction and an X-axis table motor **652** for driving the X-axis slider **64** supported on the X-axis drive transmitter **651** in the X direction. The X-axis drive transmitter **651** includes a ball screw to be driven by the X-axis table motor **652**, and the X-axis slider **64** is attached to a nut of the ball screw. However, a specific configuration of the X-axis driver **65** is not limited to this example and may be a linear motor.

[0079] Furthermore, the XY θ drive table **6** includes a θ -axis table motor **66** mounted on the X-axis slider **64**. This θ -axis table motor **66** drives the chuck stage **3** in the θ direction with respect to the X-axis slider **64**.

[0080] Such an XY θ drive table **6** can drive the chuck stage **3** in the Y direction by the Y-axis table motor **632**, drive the chuck stage **3** in the X direction by the X-axis table motor **652** and drive the chuck stage **3** in the θ direction by the θ -axis table motor **66**.

[0081] Further, the laser processing apparatus **1** is provided with a laser processing part **7** for executing a laser processing for the semiconductor substrate W held on the chuck stage **3**. The laser processing part **7** includes a processing head **71** facing the semiconductor substrate W held on the chuck stage **3** from above. The processing head **71** includes a laser light source **72** for generating a laser beam B having a predetermined frequency and an optical system **73** (a lens, a diaphragm and the like) for irradiating the laser beam B emitted from the laser light source **72** to the

semiconductor substrate W. This processing head **71** has a predetermined laser irradiation position Lb and faces the laser irradiation position Lb from above in the Z direction. The processing head **71** condenses the laser beam B emitted from the laser light source **72** on the laser irradiation position Lb by the optical system **73**, thereby forming a modified layer in a part of the semiconductor substrate W overlapping the laser irradiation position Lb.

[0082] Further, the laser processing part **7** includes a Z-axis slider **78** for supporting the processing head **71** and a Z-axis driver **79** for driving the Z-axis slider **78** in the Z direction. The Z-axis driver **79** is a single-axis robot mounted on the base plate and includes a Z-axis drive transmitter **791** for supporting the Z-axis slider **78** movably in the Z direction and a Z-axis head motor **792** for driving the Z-axis slider **78** supported on the Z-axis drive transmitter **791** in the Z direction. The Z-axis drive transmitter **791** includes a ball screw to be driven by the Z-axis head motor **792**, and the Z-axis slider **78** is attached to a nut of the ball screw. However, a specific configuration of the Z-axis driver **79** is not limited to this example and may be a linear motor. Such a Z-axis driver **79** moves the laser irradiation position Lb of an infrared camera **81** in the Z direction by driving the Z-axis slider **78** supported on the Z-axis drive transmitter **791** by the Z-axis head motor **792** to move the processing head **71** supported on the Z-axis slider **78** in the Z direction.

[0083] Further, the laser processing apparatus **1** is provided with imaging parts **8** for imaging the semiconductor substrate W held on the chuck stage **3**. Particularly, two imaging parts **8** are arranged across the laser processing part **7** in the X direction. In distinguishing these two imaging parts **8**, the imaging part **8** on the (+X) side of the laser processing part **7** is referred to as the imaging part **8A** and the imaging part **8** on the (−X) side of the laser processing part **7** is referred to as the imaging part **8B**. In this way, the imaging part **8A**, the laser processing part **7** and the imaging part **8B** are arrayed in the X direction. Note that each of the imaging parts **8A**, **8B** has a common basic configuration. Therefore, components common to the imaging parts **8A**, **8B** are described without being distinguished.

[0084] The imaging part **8** includes the infrared camera **81** facing the semiconductor substrate W held on the chuck stage **3** from above. This infrared camera **81** has a predetermined imaging range Ri (in other words, a field of view) and faces this imaging range Ri from above in the Z direction. The infrared camera **81** images the imaging range Ri by detecting infrared rays emitted from the imaging range Ri and obtains an image of the imaging range Ri.

[0085] Further, the imaging part **8** includes a Z-axis slider **88** for supporting the infrared camera **81** and a Z-axis driver **89** for driving the Z-axis slider **88** in the Z direction. The Z-axis driver **89** is a single-axis robot mounted on the base plate and includes a Z-axis drive transmitter **891** for supporting the Z-axis driver **88** movably in the Z direction and a Z-axis camera motor **892** for driving the Z-axis slider **88** supported on the Z-axis drive transmitter **891** in the Z direction. The Z-axis drive transmitter **891** includes a ball screw to be driven by the Z-axis camera motor **892**, and the Z-axis slider **88** is attached to a nut of the ball screw. However, a specific configuration of the Z-axis driver **89** is not limited to this example and may be a linear motor. Such a Z-axis driver **89** moves the imaging range Ri of the infrared camera **81** in the Z direction by driving the Z-axis slider **88** supported on the Z-axis drive transmitter **891** by the Z-axis camera motor **892** to move the infrared camera **81** supported on the Z-axis slider **88** in the Z direction.

[0086] Note that the infrared camera **81** of the imaging part **8A** and the infrared camera **81** of the imaging part **8B** have mutually different resolutions. Specifically, the infrared camera **81** of the imaging part **8A** has a higher resolution, i.e. a narrower field of view, than the infrared camera **81** of the imaging part **8B**. However, the resolutions of the infrared cameras **81** need not be different between the imaging parts **8A** and **8B**, and these infrared cameras **81** may have the same resolution. Further, in this example, centers of each of the imaging range Ri of the imaging part **8A**, the laser irradiation position Lb of the processing head **71** and the imaging range Ri of the imaging part **8B** are arranged in parallel to the X direction. However, these need not necessarily be parallel to the X direction, and the imaging range Ri of the imaging part **8A** only have to be located on the (+X) side

and the imaging range Ri of the imaging part 8B only have to be located on the (-X) side with respect to the laser irradiation position Lb of the processing head 71.

[0087] FIG. 3 is a block diagram showing the electrical configuration of the laser processing apparatus of FIG. 1. As shown in FIG. 3, the laser processing apparatus 1 is provided with a control unit 100 for controlling the components shown in FIGS. 1 and 2. The control unit 100 includes a handling control calculator 110 in charge of controlling a substrate conveying system (substrate storage part 2, Y-axis conveying mechanism 4 and XZ-axis conveying mechanism 5) relating to the conveyance of the semiconductor substrate W in the laser processing apparatus 1 and a laser processing control calculator 120 in charge of controlling a laser processing system (chuck stage 3, XYθ drive table 6, laser processing part 7 and imaging parts 8) relating to laser processing for the semiconductor substrate W.

[0088] Further, the control unit 100 includes a cassette controller 111 for controlling inserting and withdrawing operations of the semiconductor substrate W into and from the substrate storage cassette 21 in response to a command from the handling control calculator 110. This cassette controller 111 adjusts the position in the Z direction of the substrate storage cassette 21 by controlling the Z-axis cassette motor 272 and adjusts the position in the Y direction of the lift hand 41 by controlling the Y-axis lift hand motor 452.

[0089] Further, the control unit 100 includes a hand controller 112 for controlling a conveying operation of the semiconductor substrate W by the suction hand 51 in response to a command from the handling control calculator 110. The hand controller 112 adjusts the position in the X direction of the suction hand 51 by controlling the X-axis suction hand motor 552 and adjusts the position in the Z direction of the suction hand 51 by controlling the Z-axis suction hand motor 582. Further, the hand controller 112 controls a suction pump 591 for sucking the suction holes open in the bottom surface 513 of the annular suction member 512 of the suction hand 51. That is, the hand controller 112 sucks the ring frame Fr by the suction hand 51 by supplying a negative pressure to the suction holes by the suction pump 591 and separates the ring frame Fr from the suction hand 51 by stopping the supply of the negative pressure to the suction holes by the suction pump 591.

[0090] Further, the control unit 100 includes a stage controller 121 for controlling a substrate fixing operation by the chuck stage 3 and the drive of the chuck stage 3 in response to a command from the laser processing control calculator 120. The stage controller 121 adjusts the positions in the X direction, the Y direction and the θ direction of the chuck stage 3 by controlling each of the X-axis table motor 652, the Y-axis table motor 632 and the θ-axis table motor 66. Further, the stage controller 121 executes fixation of the ring frame Fr to the suction plate 31 and releasing the fixation by a clamber driver 691 by controlling the clamber driver 691 for driving the clammers 32. Furthermore, the stage controller 121 controls a suction pump 692 for sucking the suction holes open in the upper surface 311 of the suction plate 31. That is, the stage controller 121 sucks the tape E by the suction plate 31 by supplying the negative pressure to the suction holes by the suction pump 692 and releases the suction of the tape E by the suction plate 31 by stopping the supply of the negative pressure to the suction holes by the suction pump 692.

[0091] Further, the control unit 100 includes a camera controller 122A for controlling the imaging part 8A and a camera controller 122B for controlling the imaging part 8B. These hand controllers 112A, 112B execute the following controls for the infrared cameras 81 and the Z-axis camera motors 892 of the imaging parts 8A, 8B which are these targets respectively. That is, each of the camera controllers 122A, 122B causes the infrared camera 81 to image the semiconductor substrate W to obtain an image of the semiconductor substrate W, and drives the infrared camera 81 in the Z direction by the Z-axis camera motor 892 to adjust a distance in the Z direction from the infrared camera 81 to the semiconductor substrate W.

[0092] Further, the control unit 100 includes a processing head controller 123 for controlling the laser processing part 7. The processing head controller 123 drives the laser light source 72 to emit a laser beam B from the laser light source 72 and drives the processing head 71 in the Z direction by

the Z-axis head motor **792** to adjust a distance in the Z direction from the processing head **71** to the semiconductor substrate **W**. Further, the processing head **71** includes a height detector **74** for detecting a height (distance in the Z direction) from the semiconductor substrate **W**. This height detector **74** is a so-called distance sensor. Furthermore, the optical system **73** of the processing head **71** includes a focusing mechanism **75**. The focusing mechanism **75** adjusts a position to which the laser beam **B** is condensed by displacing a focus of the optical system **73** in the Z direction. Particularly, the processing head controller **123** condenses the laser beam **B** at a predetermined position inside the semiconductor substrate **W** by controlling the focusing mechanism **75** based on the height of the processing head **71** from the semiconductor substrate **W** detected by the height detector **74**.

[0093] Note that each function of the control unit **100** described above can be realized by a processor such as a CPU (Central Processing Unit), an FPGA (Field Programmable Gate Array), etc.

[0094] Further, the control unit **100** includes a storage **190**, which is a storage device such as a HDD (Hard Disk Drive) or an SSD (Solid State Drive). A laser processing program **191** for specifying later-described operations performed in the laser processing apparatus **1** for laser processing the semiconductor substrate **W** is stored in this storage **190**. That is, the control unit **100** executes each control to be described later using FIGS. **4** to **22C** by implementing the laser processing program **191**. Note that the laser processing program **191** is provided by a recording medium **192** external of the laser processing apparatus **1**, and the control unit **100** (computer) reads the laser processing program **191** recorded in the recording medium **192** and stores the laser processing program **191** in the storage **190**. Examples of such a recording medium **192** includes a USB (Universal Serial Bus) memory, a storage device of an external computer and the like.

[0095] FIG. **4** is a flow chart showing an example of a method for producing a laser processed substrate, for which the laser processing has been already performed. The flow chart of FIG. **4** is carried out in accordance with a control of the control unit **100** based on the laser processing program **191**. In Step **S101**, the lift hand **41** takes out the ring frame **Fr** from the substrate storage cassette **21** to the substrate transfer region **Aw**. In Step **S102**, the suction hand **51** in the substrate transfer region **Aw** transfers the ring frame **Fr** from the lift hand **41** to the chuck stage **3**. In this way, the semiconductor substrate **W** held by the ring frame **Fr** is taken out from the substrate storage cassette **21** to the substrate transfer region **Aw** and then transferred to the chuck stage **3** from the substrate transfer region **Aw**. Specifically, the take-out of the ring frame of FIG. **5** is performed in Step **S101**, and the transfer of the ring frame of FIG. **6** is performed in Step **S102**.

[0096] FIG. **5** is a flow chart showing an example of the take-out of the ring frame, FIG. **6** is a flow chart showing an example of the transfer of the ring frame, and FIGS. **7A** to **7E** are plan views schematically showing an example of an operation performed in accordance with the flow charts of FIGS. **5** and **6**.

[0097] In Step **S201** of FIG. **5**, the control unit **100** confirms whether or not the lift hand **41** is empty, i.e. whether or not the ring frame **Fr** is not placed on the lift hand **41**. Whether or not the lift hand **41** is empty can be confirmed based on a history or the like of operations performed by the lift hand **41**. The flow chart of FIG. **5** is finished if the lift hand **41** is not empty ("NO" in Step **S201**), whereas advance is made to Step **S201** if the lift hand **41** is empty ("YES" in Step **S201**).

[0098] In Step **S202**, the control unit **100** confirms whether or not at least a part of the lift hand **41** is located in the substrate storage cassette **21**, in other words, located on the inner side (i.e. (+Y) side) of the substrate storage cassette **21** than the opening **23** of the substrate storage cassette **21**. Whether or not at least a part of the lift hand **41** is located in the substrate storage cassette **21** can be confirmed, for example, based on the position of the lift hand **41** indicated by an output of an encoder of the Y-axis lift hand motor **452** for driving the lift hand **41** in the Y direction. Advance is made to Step **S204** without performing Step **S203** if the lift hand **41** is retracted toward the (-Y) side from the substrate storage cassette **21** ("NO" in Step **S202**), whereas advance is made to Step

S203 if a part of the lift hand **41** is located in the substrate storage cassette **21** (“YES” in Step **S202**). In Step **S203**, the control unit **100** withdraws the lift hand **41** from the substrate storage cassette **21** toward the (−Y) side and retracts the lift hand **41** toward the (−Y) side of the substrate storage cassette **21** by driving the lift hand **41** toward the (−Y) side by the Y-axis lift hand motor **452**.

[0099] In Step **S204**, the control unit **100** positions the slot **25** storing the ring frame Fr to be taken out at a position higher than the substrate insertion height **211** by a predetermined height by driving the substrate storage cassette **21** in the Z direction by the Z-axis cassette motor **272**. This predetermined height is shorter than an interval between the slots **25** adjacent in the Z direction. In this way, the bottom surface of the ring frame Fr to be taken out is adjusted to a position higher than the lift hand **41** by the predetermined height.

[0100] In Step **S205**, as shown in FIG. 7A, the control unit **100** inserts the lift hand **41** into the substrate storage cassette **21** by driving the lift hand **41** toward the (+Y) side by the Y-axis lift hand motor **452**. In this way, the lift hand **41** faces the ring frame Fr to be taken out across a gap from below.

[0101] In Step **S206**, the control unit **100** lowers the substrate storage cassette **21** in the Z direction by the Z-axis cassette motor **272**. Therefore, the ring frame Fr to be taken out is placed on the lift hand **41** and moves upward with respect to the slot **25** (i.e. the pair of supporting projections **24** specifying the slot **25**).

[0102] In Step **S207**, the control unit **100** withdraws the lift hand **41** to the substrate transfer region Aw provided outside the substrate storage cassette **21** by driving the lift hand **41** toward the (−Y) side by the Y-axis lift hand motor **452**. In this way, as shown in FIG. 7B, the ring frame Fr placed on the lift hand **41** is located in the substrate transfer region Aw.

[0103] In Step **S301** of FIG. 6, the control unit **100** causes the suction hand **51** to face the ring frame Fr supported on the lift hand **41** in the substrate transfer region Aw from above by adjusting the position in the X direction of the suction hand **51** by the X-axis suction hand motor **552** as shown in FIG. 7C. At this time, the control unit **100** adjusts the suction hand **51** to a position higher than the ring frame Fr by adjusting the height of the suction hand **51** by the Z-axis suction hand motor **582**. Therefore, the suction hand **51** faces the ring frame Fr across a gap.

[0104] In Step **S302**, the control unit **100** brings the bottom surface **513** of the suction hand **51** into contact with the upper surface of the ring frame Fr by lowering the suction hand **51** facing the ring frame Fr by the Z-axis drive transmitter **581**. In Step **S303**, the control unit **100** causes the suction pump **591** to generate a negative pressure in the suction holes provided in the bottom surface **513** of the suction hand **51** and the suction hand **51** sucks the ring frame Fr by this negative pressure. In this way, the ring frame Fr is held by the suction hand **51**. In Step **S304**, the control unit **100** raises the suction hand **51** by the Z-axis suction hand motor **582**. In this way, the suction hand **51** lifts up the ring frame Fr from the lift hand **41**.

[0105] In Step **S305**, the control unit **100** causes the suction hand **51** to face from above the chuck stage **3** as a transfer destination of the ring frame Fr by driving the suction hand **51** toward the (+X) side by the X-axis suction hand motor **552** as shown in FIG. 7D. At this time, the control unit **100** adjusts the ring frame Fr held by the suction hand **51** to a position higher than the chuck stage **3** by adjusting the height of the suction hand **51** by the Z-axis suction hand motor **582**. Therefore, the ring frame Fr held by the suction hand **51** faces the chuck stage **3** across a gap.

[0106] In Step **S306**, the control unit **100** places the ring frame Fr (and the tape E) held by the suction hand **51** on the suction plate **31** of the chuck stage **3** by lowering the suction hand **51** by the Z-axis suction hand motor **582**. In Step **S307**, the control unit **100** releases the suction of the ring frame Fr by the suction hand **51** by stopping the suction pump **591**.

[0107] In Step **S308**, the control unit **100** confirms whether or not the transfer destination of the ring frame Fr is the chuck stage **3**. For example, if the transfer destination of the ring frame Fr is the lift hand **41** as in Step **S104** to be described later, “NO” is determined in Step **S308** and the flow

chart of FIG. 6 is finished. Here, since the transfer destination of the ring frame Fr is the chuck stage 3, "YES" is determined in Step S308 and advance is made to Step S309.

[0108] In Step S309, the control unit 100 sandwiches the ring frame Fr placed on the suction plate 31 of the chuck stage 3 between the clampers 32 and the suction plate 31 to clamp the ring frame Fr by driving the clampers 32 by the clamper driver 691. Further, in Step S310, the control unit 100 causes the suction pump 591 to generate a negative pressure in the suction holes provided in the upper surface 311 of the suction plate 31 and the suction plate 31 sucks the tape E bonded to the ring frame Fr by this negative pressure. In this way, the ring frame Fr is held by the chuck stage 3. In Step S311, the control unit 100 raises the suction hand 51 by the Z-axis suction hand motor 582. This causes the suction hand 51 to retract upward from the ring frame Fr held on the chuck stage 3. In this way, as shown in FIG. 7E, the transfer of the ring frame Fr from the substrate storage cassette 21 to the chuck stage 3 is completed (Steps S101, S102 of FIG. 4).

[0109] In Step S103 of FIG. 4, substrate processing is performed to process the semiconductor substrate W held on the chuck stage 3 by the laser beam B and the laser beam B is irradiated to the plurality of planned dividing lines provided on the semiconductor substrate W. This substrate processing is described in detail later.

[0110] If the substrate processing is completed, Steps S104, S105 are performed. In Step S104, the suction hand 51 transfers the ring frame Fr from the chuck stage 3 to the lift hand 41 in the substrate transfer region Aw. In Step S105, the lift hand 41 stores the ring frame Fr into the substrate storage cassette 21 from the substrate transfer region Aw. In this way, the semiconductor substrate W held on the ring frame Fr is stored into the substrate storage cassette 21 from the substrate transfer region Aw after being transferred from the chuck stage 3 to the substrate transfer region Aw. Specifically, the transfer of the ring frame of FIG. 6 is performed in Step S104, the storage of the ring frame of FIG. 8 is performed in Step S105 and an operation opposite to the one shown in FIGS. 7A to 7E described above is performed. Here, FIG. 8 is a flow chart showing an example of the storage of the ring frame.

[0111] Since the operation of FIG. 6 performed in Step S104 is similar to the above operation of FIG. 6 performed in Step S102, description is centered on differences from the above operation and the description of common operation parts is omitted as appropriate. In Step S301 of FIG. 6, the control unit 100 causes the suction hand 51 to face the ring frame Fr placed on the chuck stage 3 from above by adjusting the position in the X direction of the suction hand 51 by the X-axis suction hand motor 552. Then, the control unit 100 lowers the suction hand 51 to the ring frame Fr (Step S302) and causes the suction hand 51 to suck the ring frame Fr (Step S303). Subsequently, the control unit 100 raises the suction hand 51 (Step S304). In this way, the suction hand 51 lifts up the ring frame Fr from the chuck stage 3.

[0112] In Step S305, the control unit 100 drives the suction hand 51 toward the (-X) side by the X-axis suction hand motor 552. At this time, the lift hand 41 is waiting on standby in the substrate transfer region Aw. In this way, the suction hand 51 faces from above the lift hand 41 in the substrate transfer region Aw as a transfer destination of the ring frame Fr. Then, the control unit 100 places the ring frame Fr held by the suction hand 51 on the lift hand 41 by lowering the suction hand 51 by the Z-axis suction hand motor 582 (Step S306). Then, the control unit 100 releases the suction of the ring frame Fr by the suction hand 51 by stopping the suction pump 591 (Step S307). In Step S308, the control unit 100 confirms whether or not the transfer destination of the ring frame Fr is the chuck stage 3. Since the transfer destination of the ring frame Fr is not the chuck stage 3, but the lift hand 41 here, "NO" is determined in Step S308 and the flow chart of FIG. 6 is finished.

[0113] In Step S401 of FIG. 8, the control unit 100 confirms whether or not the ring frame Fr has been placed on the lift hand 41. The placement of the ring frame Fr on the lift hand 41 can be confirmed, for example, based on a history of operations of the suction hand 51 for placing the ring frame Fr. If the placement of the ring frame Fr on the lift hand 41 is confirmed ("YES" in Step S401), the control unit 100 confirms whether or not at least a part of the lift hand 41 is located in

the substrate storage cassette **21** (Step **S402**) as in Step **S202** described above. Advance is made to Step **S404** without performing Step **S403** if the lift hand **41** is retracted toward the (−Y) side from the substrate storage cassette **21** (“NO” in Step **S402**), whereas advance is made to Step **S403** if a part of the lift hand **41** is located in the substrate storage cassette **21** (“YES” in Step **S402**). In Step **S403**, the control unit **100** withdraws the lift hand **41** toward the (−Y) side of the substrate storage cassette **21** and retracts the lift hand **41** toward the (−Y) side from the substrate storage cassette **21** by driving the lift hand **41** toward the (−Y) side by the Y-axis lift hand motor **452**.

[0114] In Step **S404**, the control unit **100** positions the slot **25** (in other words, the pair of supporting projections **24** specifying the slot **25**) to which the ring frame Fr is to be stored at a position lower than the substrate insertion height **211** by a predetermined height by driving the substrate storage cassette **21** in the Z direction by the Z-axis cassette motor **272**. In this way, the slot **25** for storage is adjusted to the position lower than the bottom surface of the ring frame Fr supported on the lift hand **41** by the predetermined height.

[0115] In Step **S405**, the control unit **100** inserts the lift hand **41** into the substrate storage cassette **21** by driving the lift hand **41** toward the (+Y) side by the Y-axis lift hand motor **452**. In this way, the pair of supporting projections **24** specifying the slot **25** for storage face the ring frame Fr supported on the lift hand **41** across a gap from below.

[0116] In Step **S406**, the control unit **100** raises the substrate storage cassette **21** in the Z direction by the Z-axis cassette motor **272**. In this way, the ring frame Fr is placed on the pair of supporting projections **24** specifying the slot **25** for storage and raised with respect to the lift hand **41**. In Step **S407**, the control unit **100** withdraws the lift hand **41** to the outside of the substrate storage cassette **21** by driving the lift hand **41** toward the (−Y) side by the Y-axis lift hand motor **452**.

[0117] Note that, in taking out or storing the ring frame Fr from or into the substrate storage cassette **21**, a ring frame alignment of aligning the ring frame Fr with respect to the lift hand **41** can be performed as appropriate. FIG. **9** is a flow chart showing an example of the ring frame alignment, and FIG. **10** shows plan views schematically showing an example of an operation performed in the ring frame alignment. Note that the flow chart of FIG. **9** is performed by a control of the control unit **100**.

[0118] In FIG. **10**, members (alignment projections **413** and the like) below the suction hand **51** are shown through the suction hand **51**. That is, in this example, the lift hand **41** includes a plurality of the alignment projections **413** projecting upward from the base part **41**. The plurality of these alignment projections **413** correspond to the plurality of slits Fs of the ring frame Fr. The ring frame alignment is performed using the alignment projections **413** and the slits Fs.

[0119] In this ring frame alignment, the ring frame Fr on the lift hand **41** is sucked by the suction hand **51** (Step **S501**). Then, the suction hand **51** holding the ring frame Fr is raised to separate the ring frame Fr upward from the lift hand **41** (Step **S502**). At this time, a separation height of the ring frame Fr from the lift hand **41** is so adjusted that the ring frame Fr is located at a height between the lower and upper ends of the alignment projections **413** in the Z direction.

[0120] In Step **S503**, an XYθ floating mechanism **561** built in the Z-axis slider **56** is turned on. This XYθ floating mechanism **561** selectively takes a floating state for floatingly supporting the suction hand **51** and a locking state for fixedly supporting the suction hand **51**. Here, the floating support means the support of the suction hand **51** in a state where the suction hand **51** can move in the X direction, the Y direction and the θ direction with respect to the XYθ floating mechanism **561**, and the fixed support means the support of the suction hand **51** in a state where the suction hand **51** is fixed to the XYθ floating mechanism **561**. If the XYθ floating mechanism **561** is turned on in Step **S503**, the XYθ floating mechanism **561** floatingly supports the suction hand **51** and the suction hand **51** becomes movable in the X direction, the Y direction and the θ direction with respect to the XYθ floating mechanism **561**.

[0121] In Step **S504**, the lift hand **41** moves in the Y direction and the alignment projections **413** of the lift hand **41** are brought into contact with the peripheral edge of the ring frame Fr held by the

suction hand **51**. At this time, the suction hand **51** moves with respect to the XYθ floating mechanism **561** such that the alignment projections **413** follow the peripheral edge of the ring frame Fr. As a result, as shown in field of Step S504 of FIG. **10**, the respective alignment projections **413** of the lift hand **41** are engaged with the respective slits Fs of the ring frame Fr and the ring frame Fr is positioned with respect to the lift hand **41**.

[0122] In Step S505, the XYθ floating mechanism **561** is locked. In this way, the suction hand **51** is fixedly supported by the XYθ floating mechanism **561**. Then, in Step S506, the suction of the ring frame Fr by the suction hand **51** is released and the ring frame Fr is placed on the lift hand **41**. In Step S507, the XYθ floating mechanism **561** is turned off, and the suction hand **51** is supported by the Z-axis slider **56** while being fixed to the Z-axis slider **56**. In this way, the ring frame Fr can be positioned with respect to the lift hand **41** (ring frame alignment).

[0123] Next, the substrate processing is described in detail. FIG. **11** is a flow chart showing an example of the substrate processing, and FIG. **12** shows plan views schematically showing an example of an operation performed in accordance with the flow chart of FIG. **11**. The flow chart of FIG. **11** is performed by a control of the control unit **100**.

[0124] In Step S601 of the substrate processing of FIG. **11**, calibration is performed to obtain a plane of the upper surface (back surface) of the semiconductor substrate W to be processed. FIG. **13A** is a flow chart showing an example of the calibration, FIG. **13B** is a flow chart showing an example of stage plane specification performed in the calibration of FIG. **13A**, and FIG. **13C** is a flow chart showing an example of substrate plane specification performed in the calibration of FIG. **13A**. Note that, in the calibration of FIG. **13A**, the suction plate **31** or the semiconductor substrate W is imaged as appropriate. In this description, it is assumed that imaging is performed by the imaging part **8B**. However, the following operation can also be similarly performed even if imaging is performed by the imaging part **8A**.

[0125] In Step S701 of the calibration of FIG. **13A**, the stage plane specification (FIG. **13B**) is performed. As shown in FIG. **13B**, in the stage plane specification, a count value I for discriminating a plurality of (three) imaging points Ps(I) provided on the upper surface **311** of the suction plate **31** of the chuck stage **3** is reset to zero (Step S801), and the count value I is incremented by 1 (Step S802). The imaging point Ps(I) is, for example, a mark having a predetermined pattern.

[0126] In Step S803, the control unit **100** causes the imaging point Ps(I) to face the infrared camera **81** from below by adjusting the position of the chuck stage **3** by the XYθ drive table **6**. In this way, the imaging point Ps(I) falls within a field of view of the infrared camera **81**. In Step S803, the infrared camera **81** images this imaging point Ps(I) and obtains an image showing the imaging point Ps(I). In Step S804, the control unit **100** confirms whether or not the predetermined pattern of the imaging point Ps(I) can be detected from the image by an image processing such as pattern matching.

[0127] If a focus of the infrared camera **81** deviates from the imaging point Ps(I) and the predetermined pattern cannot be detected from the image (“NO” in Step S804), the control unit **100** changes a distance of the infrared camera **81** to the imaging point Ps(I) in the Z direction by driving the infrared camera **81** in the Z direction by the Z-axis camera motor **892** (Step S805). In this way, the focus of the infrared camera **81** is changed in the Z direction. Steps S803 to S805 are repeated until the focus of the infrared camera **81** coincides with the imaging point Ps(I) and the predetermined pattern is detected (“YES” in Step S804).

[0128] In Step S806, the control unit **100** calculates the position (X, Y, Z) of the imaging point Ps(I) based on the predetermined pattern detected from the image obtained by imaging the imaging point Ps(I). X- and Y-coordinates of the imaging point Ps(I) are calculated based on the position of the predetermined pattern included in the image. A Z-coordinate of the imaging point Ps(I) is calculated based on the position of the infrared camera **81** in the Z direction when the image, from which the predetermined pattern could be detected, was imaged.

[0129] In Step S807, it is confirmed whether or not the count value I has reached 2, i.e. the positions (X, Y, Z) of two imaging points Ps(1), Ps(2) have been obtained. If the count value I is less than 2 ("NO" in Step S807), return is made to Step S802 and Steps S802 to S806 are performed. If the count value I is 2 ("YES" in Step S807), advance is made to Step S808.

[0130] In Step S808, a rotation angle θ_a for rotating the chuck stage 3 in the θ direction is so calculated that a straight line passing through the two imaging points Ps(1) and Ps(2) is horizontal. If a difference from the current rotation angle of the suction plane 31 (difference between an actual rotation angle and the rotation angle θ_a) is not zero ("NO" in Step S809), the chuck stage 3 is rotated by the rotation angle θ_a (Step S810) and return is made to Step S801. In this way, Steps S801 to S809 are performed.

[0131] If the difference from the current rotation angle of the suction plane 31 (difference between the actual rotation angle and the rotation angle θ_a) is zero ("YES" in Step S809), advance is made to Step S811. In Step S811, the control unit 100 images the imaging point Ps(3) by the infrared camera 81 and obtains an image showing the imaging point Ps(3) in the same manner as in Step S803. In Step S812, the control unit 100 confirms whether or not a predetermined pattern included in the imaging point Ps(3) can be detected from this image by an image processing such as pattern matching.

[0132] If the predetermined pattern cannot be detected from the image ("NO" in Step S812), the control unit 100 changes a distance of the infrared camera 81 to the imaging point Ps(3) in the Z direction by driving the infrared camera 81 in the Z direction by the Z-axis camera motor 892 (Step S813). Steps S811 to S813 are repeated until the predetermined pattern is detected ("YES" in Step S812).

[0133] If the predetermined pattern can be detected in Step S812 (YES), the control unit 100 calculates the position (X, Y, Z) of the imaging point Ps(3) based on the predetermined pattern detected from the image obtained by imaging the imaging point Ps(3) (Step S814). In this way, the position (X, Y, Z) of each of the three imaging points Ps(1), Ps(2) and Ps(3) is obtained. In Step S815, a plane passing through these three positions (X, Y, Z) is specified as a plane representing the plane of the chuck stage 3, specifically the upper surface 311 of the suction plane 31.

[0134] In Step S702 of the calibration of FIG. 13A, the substrate plane specification (FIG. 13C) is performed. As shown in FIG. 13C, in the substrate plane specification, a count value I for discriminating a plurality of (three) imaging points Pw(I) of the semiconductor substrate W is reset to zero (Step S901), and the count value I is incremented by 1 (Step S902). The imaging points Pw(I) is, for example, an area having a predetermined pattern.

[0135] Specifically, as shown in FIG. 12, the semiconductor substrate W is demarcated in the form of a lattice by planned dividing lines S (Sa, Sb) orthogonal to each other. That is, the semiconductor substrate W is provided with a plurality of the planned dividing lines Sa parallel to each other and a plurality of the planned dividing lines Sb parallel to each other, and the planned dividing lines Sa and the planned dividing lines Sb are orthogonal to each other. In this way, a plurality of semiconductor chips C are arrayed in a lattice across the planned dividing lines Sa, Sb. In contrast, a region including an intersection of the planned dividing line Sa and the planned dividing line Sb (in other words, a point surrounded by the semiconductor chips C arranged on four corners) is set as the imaging point Pw(I). Note that, since the back surface of the semiconductor substrate W is facing upward as described above, the infrared camera 81 images the planned dividing lines Sa, Sb and the semiconductor chips C formed on the front surface of the semiconductor substrate W through the back surface of the semiconductor substrate W by infrared rays.

[0136] In Step S903, the control unit 100 causes the imaging point Pw(I) to face the infrared camera 81 from below by adjusting the position of the chuck stage 3 by the XY θ drive table 6. In this way, the imaging point Pw(I) falls within the field of view of the infrared camera 81. In Step S903, the infrared camera 81 images this imaging point Pw(I) and obtains an image showing the imaging point Pw(I). In Step S904, the control unit 100 confirms whether or not a predetermined

pattern (e.g. an intersection pattern of the planned dividing lines Sa, Sb) included in the imaging point Pw(I) can be detected from this image by an image processing such as pattern matching.

[0137] If the focus of the infrared camera **81** deviates from the imaging point Pw(I) and the predetermined pattern cannot be detected from the image (“NO” in Step S904), the control unit **100** changes a distance of the infrared camera **81** to the imaging point Pw(I) in the Z direction by driving the infrared camera **81** in the Z direction by the Z-axis camera motor **892** (Step S905). In this way, the focus of the infrared camera **81** is changed in the Z direction. Steps S903 to S905 are repeated until the focus of the infrared camera **81** coincides with the imaging point Pw(I) and the predetermined pattern is detected (“YES” in Step S904).

[0138] Note that the plane (stage plane) representing the upper surface **311** of the suction plane **31** is specified by the previously performed stage plane specification (FIG. 13B). Accordingly, a height range in which the imaging point Pw(I) of the semiconductor substrate W placed on the suction plane **31** is present can be estimated based on this stage plane. Therefore, in Step S805, the height of the infrared camera **81** is so changed that the focus of the infrared camera **81** falls within a presence range of the imaging point Pw(I) estimated from the stage plane.

[0139] In Step S906, the control unit **100** calculates the position (X, Y, Z) of the imaging point Pw(I) based on the predetermined pattern detected from the image obtained by imaging the imaging point Pw(I). X- and Y-coordinates of the imaging point Pw(I) are calculated based on the position of the predetermined pattern included in the image. AZ-coordinate of the imaging point Pw(I) is calculated based on the position of the infrared camera **81** in the Z direction when the image, from which the predetermined pattern could be detected, was imaged.

[0140] In Step S907, it is confirmed whether or not the count value I has reached 2, i.e. the positions (X, Y, Z) of two imaging points Ps(1), Ps(2) have been obtained. If the count value I is less than 2 (“NO” in Step S907), return is made to Step S902 and Steps S902 to S906 are performed. If the count value I is 2 (“YES” in Step S907), advance is made to Step S908.

[0141] In Step S908, a rotation angle θ_b for rotating the chuck stage **3** in the θ direction is so calculated based on the two imaging points Pw(1), Pw(2) that the planned dividing lines Sa are parallel to the X direction (processing direction). If a difference from the current rotation angle of the suction plane **31** (difference between an actual rotation angle and the rotation angle θ_b) is not zero (“NO” in Step S909), the chuck stage **3** is rotated by the rotation angle θ_b (Step S910) and return is made to Step S901. In this way, Steps S901 to S909 are performed.

[0142] If the difference from the current rotation angle of the suction plane **31** (difference between the actual rotation angle and the rotation angle θ_b) is zero (“YES” in Step S909), advance is made to Step S911. In Step S911, the control unit **100** images the imaging point Pw(3) by the infrared camera **81** and obtain an image showing the imaging point Pw(3) in the same manner as in Step S903. In Step S912, the control unit **100** confirms whether or not a predetermined pattern included in the imaging point Pw(3) can be detected from this image by an image processing such as pattern matching.

[0143] If the predetermined pattern cannot be detected from the image (“NO” in Step S912), the control unit **100** changes a distance of the infrared camera **81** to the imaging point Pw(3) in the Z direction by driving the infrared camera **81** in the Z direction by the Z-axis camera motor **892** (Step S913). Steps S911 to S913 are repeated until the predetermined pattern is detected (“YES” in Step S912). At this time, a range for changing the height of the infrared camera **81** is set based on the stage plane as in the aforementioned case.

[0144] If the predetermined pattern can be detected in Step S912 (YES), the control unit **100** calculates the position (X, Y, Z) of the imaging point Pw(3) based on the predetermined pattern detected from the image obtained by imaging the imaging point Pw(3) (Step S914). In this way, the position (X, Y, Z) of each of the three imaging points Pw(1), Pw(2) and Pw(3) is obtained. In Step S915, a plane passing through these three positions (X, Y, Z) is specified as a plane representing the semiconductor substrate W.

[0145] Referring back to FIG. 11, description is continued. If the semiconductor substrate W is so positioned that the planned dividing lines Sa are parallel to the X direction and the plane representing the semiconductor substrate W is specified by performing the calibration described above (Step S601), line processing (Step S602) is performed for each planned dividing line Sa. That is, each of the plurality of planned dividing lines Sa is processed by the laser beam B by performing the line processing for irradiating the laser beam B to the laser irradiation position Lb while moving the laser irradiation position Lb in the X direction along the target planned dividing line Sa while changing the target planned dividing line Sa, out of the plurality of planned dividing lines Sa. Particularly, as shown in field of Step S602 of FIG. 12, the line processing of moving the laser irradiation position Lb toward the (+X) in the X direction and the line processing of moving the laser irradiation position Lb toward the (−X) side in the X direction are alternately performed. [0146] At this time, a movement of the laser beam B toward the (+X) side with respect to the planned dividing line Sa is performed by driving the chuck stage 3 holding the semiconductor substrate W toward the (−X) side by the X-axis driver 65, and a movement of the laser beam B toward the (−X) side with respect to the planned dividing line Sa is performed by driving the chuck stage 3 holding the semiconductor substrate W toward the (+X) side by the X-axis driver 65. Further, the target planned dividing line Sa of the line processing is changed by driving the chuck stage 3 holding the semiconductor substrate W in the Y direction by the Y-axis driver 63. Further, the control unit 100 executes a control of adjusting the position in the Z direction of the processing head 71 by the Z-axis head motor 792 based on the plane representing the semiconductor substrate W specified by the calibration of Step S601. In this way, a condensing position of the laser beam B is adjusted to the inside of the semiconductor substrate W and a modified layer is formed inside the semiconductor substrate W along the planned dividing lines Sa.

[0147] If the line processing for each of the plurality of planned dividing lines Sa is completed in this way (Step S602), the chuck stage 3 holding the semiconductor substrate W is rotated by 90° in the θ direction by the θ -axis table motor 66. In this way, a switch is performed from a state where the plurality of planned dividing lines Sa, to which the line processing had been performed, are positioned in parallel to the X direction (field of “S602_e” of FIG. 12) to a state where the plurality of planned dividing lines Sb are positioned in parallel to the X direction (field of “S603” of FIG. 12).

[0148] In Step S604, the calibration is performed as in Step S601 described above. Further, in Step S605, the line processing is performed for each of the plurality of planned dividing lines Sb as in Step S602 described above.

[0149] FIG. 14 is a flow chart showing a basic process of the line processing for each planned dividing line, and FIG. 15A shows charts schematically showing a first example of an operation performed in accordance with the flow chart of FIG. 14. In FIG. 15A, a trace of the laser irradiation position Lb relatively moving with respect to the semiconductor substrate W is shown by dotted line, and virtual straight lines Sv1, Sv2, Sv3 extended in parallel to the X direction along the planned dividing lines S1, S2, S3 between both outer sides of the planned dividing lines S1, S2, S3 are shown by one-dot chain line. Note that dotted lines representing the trace of the laser irradiation position Lb are preferentially shown in parts where the trace of the laser irradiation position Lb and the virtual straight lines Sv1, Sv2, Sv3 overlap.

[0150] In the example shown in FIG. 15A, the flow chart of FIG. 14 is started from a state where the laser irradiation position Lb is stopped at a position Pb1 on the (−X) side of the semiconductor substrate W in the X direction. This position Pb1 is a position provided on the virtual straight line Sv1 along the planned dividing line S1, in other words, facing the planned dividing line S1 from the X direction. However, the position of the laser irradiation position Lb when the flow chart of FIG. 14 is started is not limited to this example and can be changed as appropriate.

[0151] In Step S1001, the laser irradiation position Lb stopped at the position Pb1 starts to accelerate toward the (+X) side in the X direction and moves in parallel to the X direction.

Thereby, the laser irradiation position L_b moves toward the (+X) side along the virtual straight line Sv1. If a velocity V_x of the laser irradiation position L_b increases to a processing velocity V_{xd} by the time the laser irradiation position L_b reaches an end of the semiconductor substrate W on the (-X) side, the laser irradiation position L_b moves at the constant processing velocity V_{xd} toward the (+X) side in the X direction (Step S1002).

[0152] Further, the laser light source 72 is turned on and the irradiation of the laser beam B to the laser irradiation position L_b from the processing head 71 is started in accordance with a timing at which the laser irradiation position L_b reaches the end of the semiconductor substrate W on the (-X) side (Step S1003). Further, the laser light source 72 is turned off and the irradiation of the laser beam B to the laser irradiation position L_b from the processing head 71 is finished in accordance with a timing at which the laser irradiation position L_b reaches an end of the semiconductor substrate W on the (+X) side (Step S1004). In this way, during a period from Step S1003 to Step S1004, the laser processing is performed for the planned dividing line S1 by irradiating the laser beam B to the laser irradiation position L_b while moving the laser irradiation position L_b toward the (+X) side along the planned dividing line S1 (line processing).

[0153] If the laser irradiation position L_b passes through the planned dividing line S1 toward the (+X) side, the laser irradiation position L_b starts to decelerate toward the (+X) side in the X direction (Step S1005) and stops at a position Pb2 on the (+X) side of the semiconductor substrate W in the X direction (Step S1006). This position Pb2 is a position provided on the virtual straight line Sv2 adjacent to the virtual straight line Sv1 in the Y direction, in other words, facing the planned dividing line S2 from the X direction. That is, in Steps S1005 to S1006, the laser irradiation position L_b moves in the Y direction from the virtual straight line Sv1 to the virtual straight line Sv2 in parallel with deceleration in the X direction.

[0154] A positional relationship of the imaging ranges R_i (FIG. 1) of the imaging parts 8A, 8B and the laser irradiation position L_b of the processing head 71 is fixed. Thus, in Steps S1001 to S1006, the imaging ranges R_i also relatively move with respect to the semiconductor substrate W as the laser irradiation position L_b relatively moves with respect to the semiconductor substrate W. With the laser irradiation position L_b stopped at the position Pb2, the imaging range R_i of the imaging part 8B stops at a position including at least an imaging point $P_w(S2)$. This imaging point $P_w(S2)$ is an intersection where the planned dividing line S2 and the planned dividing line S orthogonal to the line S2 intersect in the semiconductor substrate W. Accordingly, in Step S1006, the control unit 100 causes the imaging part 8B to image the imaging range R_i and obtains an image including the imaging point $P_w(S2)$. Thereby, the control unit 100 can obtain an image showing the position of the unprocessed planned dividing line S2.

[0155] In Step S1007, it is confirmed whether or not the laser processing has been completed for the plurality of planned dividing lines S parallel to the X direction. If there is any unprocessed planned dividing line S, out of these planned dividing lines S ("NO" in Step S1007), return is made to Step S1001.

[0156] In the example of FIG. 15A, in Step S1001, the laser irradiation position L_b stopped at the position Pb2 starts to accelerate toward the (-X) side in the X direction and moves in parallel to the X direction. Thereby, the laser irradiation position L_b moves toward the (-X) side along the virtual straight line Sv2. If the velocity V_x of the laser irradiation position L_b increases to the processing velocity V_{xd} by the time the laser irradiation position L_b reaches the end of the semiconductor substrate W on the (+X) side, the laser irradiation position L_b moves at the constant processing velocity V_{xd} toward the (-X) side in the X direction (Step S1002).

[0157] Here, a position where the laser irradiation position L_b having passed through the planned dividing line S1 toward the (+X) side starts to decelerate (in other words, an X-coordinate at which the constant velocity movement toward (+X) side is finished) and a position where the laser irradiation position L_b accelerating toward the (-X) side toward the planned dividing line S finishes acceleration (in other words, an X-coordinate at which the constant velocity movement

toward ($-X$) side is started) coincide. That is, the X -coordinate at which the constant velocity movement of the laser irradiation position Lb having passed through the planned dividing line Sn , to which the line processing is to be performed in the n .sup.th turn, is finished and deceleration is started and the X direction at which the laser irradiation position Lb moving toward the planned dividing line $Sn+1$, to which the line processing is to be performed in the $(n+1)$ th turn, finishes the acceleration and starts the constant velocity movement coincide.

[0158] Further, the laser light source **72** is turned on and the irradiation of the laser beam B to the laser irradiation position Lb from the processing head **71** is started in accordance with a timing at which the laser irradiation position Lb reaches the end of the semiconductor substrate W on the $(+X)$ side (Step **S1003**). Further, the laser light source **72** is turned off and the irradiation of the laser beam B to the laser irradiation position Lb from the processing head **71** is finished in accordance with a timing at which the laser irradiation position Lb reaches the end of the semiconductor substrate W on the $(-X)$ side (Step **S1004**). In this way, during a period from Step **S1003** to Step **S1004**, the laser processing is performed for the planned dividing line $S2$ by irradiating the laser beam B to the laser irradiation position Lb while moving the laser irradiation position Lb toward the $(-X)$ side along the planned dividing line $S2$ (line processing).

[0159] If the laser irradiation position Lb passes through the planned dividing line $S2$ toward the $(-X)$ side, the laser irradiation position Lb starts to decelerate toward the $(-X)$ side in the X direction (Step **S1005**) and stops at a position $Pb3$ on the $(-X)$ side of the semiconductor substrate W in the X direction (Step **S1006**). This position $Pb3$ is a position provided on the virtual straight line $Sv3$ adjacent to the virtual straight line $Sv2$ in the Y direction, in other words, facing the planned dividing line $S3$ from the X direction. That is, in Steps **S1005** to **S1006**, the laser irradiation position Lb moves in the Y direction from the virtual straight line $Sv2$ to the virtual straight line $Sv3$ in parallel with deceleration in the X direction.

[0160] With the laser irradiation position Lb stopped at the position $Pb3$, the imaging range Ri of the imaging part **8A** stops at a position including at least an imaging point $Pw(S3)$. This imaging point $Pw(S3)$ is an intersection of the planned dividing line $S3$ and the planned dividing line S orthogonal to this line $S3$ in the semiconductor substrate W . Accordingly, in Step **S1006**, the control unit **100** causes the imaging part **8A** to image the imaging range Ri and obtains an image including the imaging point $Pw(S3)$. In this way, the control unit **100** can obtain an image showing the position of the unprocessed planned dividing line $S3$. Steps **S1001** to **S1007** are repeated until it is confirmed that the line processing has been completed for the plurality of planned dividing lines S ($S1, S2, S3, \dots$) parallel to the X direction (“YES” in Step **S1007**).

[0161] Next, a velocity change of the laser irradiation position Lb is described with reference to “Velocity Change in X Direction” and “Velocity Change in Y Direction” of FIG. **15A**. Here, the velocity Vx indicates a velocity to move the laser irradiation position Lb in the X direction with respect to the semiconductor substrate W , and a velocity Vy indicates a velocity to move the laser irradiation position Lb in the Y direction with respect to the semiconductor substrate W . Further, the processing velocity Vxd indicates a velocity to move the laser irradiation position Lb at a constant velocity in the X direction along the planned dividing line S (i.e. the velocity Vx), and is represented by an absolute value regardless of the movement toward the $(+X)$ side or the movement toward the $(-X)$ side.

[0162] In a line processing period $Ts1$ (Steps **S1002** to **S1004**) during which the line processing of moving the laser beam B toward the $(+X)$ side along the planned dividing line $S1$ is performed, the laser irradiation position Lb does not move in the Y direction while moving at the constant processing velocity Vxd in the X direction. Further, in a line processing period $Ts2$ (Steps **S1002** to **S1004**) during which the line processing of moving the laser beam B toward the $(-X)$ side along the planned dividing line $S2$ is performed, the laser irradiation position Lb does not move in the Y direction while moving at the constant processing velocity Vxd in the X direction.

[0163] In a switching period Tc (Steps **S1005**, **S1006** and **S1001**) during which a switch is

performed from the line processing period Ts1 to the line processing period Ts2, the following operation is performed. That is, the X-axis driver **65** (processing-axis driver) performs reverse drive to bring the laser irradiation position Lb to the planned dividing line S2 (second processing line) by accelerating the laser irradiation position Lb toward the (-X) side (Step S1001) after decelerating and stopping the laser irradiation position Lb, which has passed through the planned dividing line S1 (first processing line) toward the (+X) side (first side), toward the (+X) side in the X direction (processing direction) (Step S1005). In parallel with this reverse drive, the Y-axis driver **63** (feeding-axis driver) moves the laser irradiation position Lb in the Y direction (feeding direction) from the virtual straight line Sv1 (first virtual straight line) extended in the X direction to the outside of the planned dividing line S1 along the planned dividing line S1 to the virtual straight line Sv2 (second virtual straight line) extended in the X direction to the outside of the planned dividing line S2 along the planned dividing line S2 (second processing line).

[0164] Particularly, the switching period Tc includes a deceleration period Td (Step S1005) for decelerating the laser irradiation position Lb in the X direction and an acceleration period Ta (Step S1001) for accelerating the laser irradiation position Lb in the X direction, and a movement of the laser irradiation position Lb in the Y direction is performed during the deceleration period Td, out of the deceleration period Td and the acceleration period Ta. Specifically, the movement of the laser irradiation position Lb in the Y direction is started after the start of the deceleration period Td, and the movement of the laser irradiation position Lb in the Y direction is finished before the end of the deceleration period Td. Further, the laser irradiation position Lb does not move in the Y direction in the acceleration period Ta.

[0165] Here, a start point of the deceleration period Td indicates a point of time at which the deceleration of the laser irradiation position Lb in the X direction is started (in other words, a decrease of the absolute value of the velocity Vx from the processing velocity Vxd is started), and an end point of the deceleration period Td indicates a point of time at which the velocity of the laser irradiation position Lb in the X direction (in other words, the velocity Vx) becomes zero. A start point of the acceleration period Ta indicates a point of time at which the acceleration of the laser irradiation position Lb in the X direction is started (in other words, an increase of the absolute value of the velocity Vx from zero is started), and an end point of the acceleration period Ta indicates a point of time at which the acceleration of the laser irradiation position Lb in the X direction is finished (in other words, a point of time at which the absolute value of the velocity Vx becomes the processing velocity Vxd).

[0166] Further, in a stop period Tt provided on the way during a transition from the acceleration period Ta to the deceleration period Td, both the velocity Vx of the laser irradiation position Lb in the X direction and the velocity Vy thereof in the Y direction become zero and the laser irradiation position Lb is stopped at the position Pb2 with respect to the semiconductor substrate W. In this stop period Tt, the imaging ranges Ri of the imaging parts **8A**, **8B** are also stopped with respect to the semiconductor substrate W. Particularly, the imaging range Ri of the imaging part **8B** is located on the (-X) side of the laser irradiation position Lb located on the (+X) side of the semiconductor substrate W and overlaps the semiconductor substrate W. Accordingly, the infrared camera **81** of the imaging part **8B** images a part of the semiconductor substrate W overlapping the imaging range Ri in the stop period Tt (Step S1006).

[0167] FIG. 15B shows charts schematically showing a second example of the operation performed in accordance with the flow chart of FIG. 14. Notation in FIG. 15B is similar to that in FIG. 15A. Also in FIG. 15B, the laser processing is performed in turn for the planned dividing lines S1, S2, S3 in accordance with the flow chart of FIG. 14 as in FIG. 15A. However, FIG. 15B differs from FIG. 15A in the operation in the switching period Tc for changing the planned dividing line S to be laser processed. Accordingly, the following description is centered on differences from FIG. 15A and common operation parts are denoted by corresponding reference signs and description is omitted as appropriate.

[0168] If the laser irradiation position Lb passes through the planned dividing line S1 toward the (+X) side as the laser processing for the planned dividing line S1 is finished, the laser irradiation position Lb starts to decelerate toward the (+X) side in the X direction (Step S1005) and stops at a position Pb2 on the (+X) side of the semiconductor substrate W in the X direction (Step S1006). This position Pb2 is a position provided on the virtual straight line Sv1. Further, with the laser irradiation position Lb stopped at the position Pb2, the imaging range Ri of the imaging part 8B stops at a position including at least an imaging point Pw(S2). Accordingly, in Step S1006, the control unit 100 causes the imaging part 8B to image the imaging range Ri and obtains an image including the imaging point Pw(S2). In this way, the control unit 100 can obtain the image showing the position of the unprocessed planned dividing line S2.

[0169] Subsequently, the laser irradiation position Lb stopped at the position Pb2 starts to accelerate toward the (−X) side in the X direction (Step S1001). If the velocity Vx of the laser irradiation position Lb increases to the processing velocity Vxd by the time the laser irradiation position Lb reaches the end of the semiconductor substrate W on the (+X) side, the laser irradiation position Lb moves at the constant processing velocity Vxd toward the (−X) side in the X direction (Step S1002). Further, in a period from the start of acceleration to the start of constant movement at the processing velocity Vxd of the laser irradiation position Lb, the laser irradiation position Lb moves in the Y direction from the virtual straight line Sv1 to the virtual straight line Sv2. That is, in Steps S1001 to S1002, the laser irradiation position Lb moves in the Y direction from the virtual straight line Sv1 to the virtual straight line Sv2 in parallel with acceleration in the X direction. In this way, the laser irradiation position Lb reaches the planned dividing line S2 and the line processing for the planned dividing line S2 can be started.

[0170] If the laser irradiation position Lb passes through the planned dividing line S2 toward the (−X) side as the laser processing for the planned dividing line S2 is finished, the laser irradiation position Lb starts to decelerate toward the (−X) side in the X direction (Step S1005) and stops at a position Pb3 on the (−X) side of the semiconductor substrate W in the X direction (Step S1006). This position Pb3 is a position provided on the virtual straight line Sv2. Further, with the laser irradiation position Lb stopped at the position Pb3, the imaging range Ri of the imaging part 8A stops at a position including at least an imaging point Pw(S3). Accordingly, in Step S1006, the control unit 100 causes the imaging part 8A to image the imaging range Ri and obtains an image including the imaging point Pw(S3). In this way, the control unit 100 can obtain the image showing the position of the unprocessed planned dividing line S3.

[0171] Next, a velocity change of the laser irradiation position Lb is described with reference to “Velocity Change in X Direction” and “Velocity Change in Y Direction” of FIG. 15B. In a line processing period Ts1 (Steps S1002 to S1004) during which the line processing of moving the laser beam B toward the (+X) side along the planned dividing line S1 is performed, the laser irradiation position Lb does not move in the Y direction while moving at the constant processing velocity Vxd in the X direction. Further, in a line processing period Ts2 (Steps S1002 to S1004) during which the line processing of moving the laser beam B toward the (−X) side along the planned dividing line S2 is performed, the laser irradiation position Lb does not move in the Y direction while moving at the constant processing velocity Vxd in the X direction.

[0172] Further, in a switching period Tc (Steps S1005, S1006 and S1001) during which a switch is performed from the line processing period Ts1 to the line processing period Ts2, the laser irradiation position Lb is moved in the Y direction (feeding direction) from the virtual straight line Sv1 to the virtual straight line Sv2 in parallel with performing the reverse drive in the X direction as in the aforementioned case. Particularly, a movement of the laser irradiation position Lb in the Y direction is performed during an acceleration period Ta, out of a deceleration period Td and the acceleration period Ta included in the switching period Tc. Specifically, the movement of the laser irradiation position Lb in the Y direction is started after the start of the acceleration period Ta, and the movement of the laser irradiation position Lb in the Y direction is finished before the end of the

acceleration period Ta. Further, the laser irradiation position Lb does not move in the Y direction in the deceleration period Td.

[0173] Further, in a stop period Tt provided on the way during a transition from the acceleration period Ta to the deceleration period Td, both the velocity Vx of the laser irradiation position Lb in the X direction and the velocity Vy thereof in the Y direction become zero and the laser irradiation position Lb is stopped at the position Pb2 with respect to the semiconductor substrate W. In this stop period Tt, the imaging ranges Ri of the imaging parts **8A**, **8B** are also stopped with respect to the semiconductor substrate W. Particularly, the imaging range Ri of the imaging part **8B** is located on the (−X) side of the laser irradiation position Lb located on the (+X) side of the semiconductor substrate W and overlaps the semiconductor substrate W. Accordingly, the infrared camera **81** of the imaging part **8B** images a part of the semiconductor substrate W overlapping the imaging range Ri in the stop period Tt (Step **S1006**).

[0174] FIG. **15C** shows charts schematically showing a third example of the operation performed in accordance with the flow chart of FIG. **14**. Notation in FIG. **15C** is similar to that in FIG. **15A**. Also in FIG. **15C**, the laser processing is performed in turn for the planned dividing lines **S1**, **S2**, **S3** in accordance with the flow chart of FIG. **14** as in FIG. **15A**. However, FIG. **15C** differs from FIG. **15A** in the operation in the switching period Tc for changing the planned dividing line S to be laser processed. Accordingly, the following description is centered on differences from FIG. **15A** and common operation parts are denoted by corresponding reference signs and description is omitted as appropriate.

[0175] If the laser irradiation position Lb passes through the planned dividing line **S1** toward the (+X) side as the laser processing for the planned dividing line **S1** is finished, the laser irradiation position Lb starts to decelerate toward the (+X) side in the X direction (Step **S1005**) and stops at a position Pb2 on the (+X) side of the semiconductor substrate W in the X direction (Step **S1006**). This position Pb2 is provided between the virtual straight lines Sv1 and Sv2 in the Y direction. That is, in Steps **S1005** to **S1006**, the laser irradiation position Lb moves in the Y direction from the virtual straight line Sv1 to the position Pb2 in parallel with deceleration in the X direction. Further, with the laser irradiation position Lb stopped at the position Pb2, the imaging range Ri of the imaging part **8B** stops at a position including at least an imaging point Pw(S2). Accordingly, in Step **S1006**, the control unit **100** causes the imaging part **8B** to image the imaging range Ri and obtains an image including the imaging point Pw(S2). In this way, the control unit **100** can obtain the image showing the position of the unprocessed planned dividing line **S2**.

[0176] Subsequently, the laser irradiation position Lb stopped at the position Pb2 starts to accelerate toward the (−X) side in the X direction (Step **S1001**). If the velocity Vx of the laser irradiation position Lb increases to the processing velocity Vxd by the time the laser irradiation position Lb reaches the end of the semiconductor substrate W on the (+X) side, the laser irradiation position Lb moves at the constant processing velocity Vxd toward the (−X) side in the X direction (Step **S1002**). Further, in a period from the start of acceleration to the start of constant movement at the processing velocity Vxd of the laser irradiation position Lb, the laser irradiation position Lb moves in the Y direction from the position Pb2 to the virtual straight line Sv2. That is, in Steps **S1001** to **S1002**, the laser irradiation position Lb moves in the Y direction from the position Pb2 to the virtual straight line Sv2 in parallel with acceleration in the X direction. In this way, the laser irradiation position Lb reaches the planned dividing line **S2** and the line processing for the planned dividing line **S2** can be started.

[0177] If the laser irradiation position Lb passes through the planned dividing line **S2** toward the (−X) side as the laser processing for the planned dividing line **S2** is finished, the laser irradiation position Lb starts to decelerate toward the (−X) side in the X direction (Step **S1005**) and stops at a position Pb3 on the (−X) side of the semiconductor substrate W in the X direction (Step **S1006**). This position Pb3 is provided between the virtual straight line Sv2 and the virtual straight line Sv3. That is, in Steps **S1005** to **S1006**, the laser irradiation position Lb moves in the Y direction from the

virtual straight line Sv2 to the position Pb3 in parallel with deceleration in the X direction. Further, with the laser irradiation position Lb stopped at the position Pb3, the imaging range Ri of the imaging part 8A stops at a position including at least the imaging point Pw(S3). Accordingly, in Step S1006, the control unit 100 causes the imaging part 8A to image the imaging range Ri and obtains an image including the imaging point Pw(S3). In this way, the control unit 100 can obtain the image showing the position of the unprocessed planned dividing line S3.

[0178] Next, a velocity change of the laser irradiation position Lb is described with reference to “Velocity Change in X Direction” and “Velocity Change in Y Direction” of FIG. 15C. In a line processing period Ts1 (Steps S1002 to S1004) during which the line processing of moving the laser beam B toward the (+X) side along the planned dividing line S1 is performed, the laser irradiation position Lb does not move in the Y direction while moving at the constant processing velocity Vxd in the X direction. Further, in a line processing period Ts2 (Steps S1002 to S1004) during which the line processing of moving the laser beam B toward the (−X) side along the planned dividing line S2 is performed, the laser irradiation position Lb does not move in the Y direction while moving at the constant processing velocity Vxd in the X direction.

[0179] Further, in a switching period Tc (Steps S1005, S1006 and S1001) during which a switch is performed from the line processing period Ts1 to the line processing period Ts2, the laser irradiation position Lb is moved in the Y direction (feeding direction) from the virtual straight line Sv1 to the virtual straight line Sv2 in parallel with performing the reverse drive in the X direction as in the aforementioned case. Particularly, this movement of the laser irradiation position Lb is performed by way of the position Pb2. That is, the laser irradiation position Lb moves in the Y direction from the virtual straight line Sv1 to the position Pb2 in a deceleration period Td, out of the deceleration period Td and an acceleration period Ta included in the switching period Tc, and moves in the Y direction from the position Pb2 to the virtual straight line Sv2 in the acceleration period Ta. Specifically, the laser irradiation position Lb starts to move from the virtual straight line Sv1 to the position Pb2 simultaneously with the start of the deceleration period Td, and reaches the position Pb2 simultaneously with the end of the deceleration period Td. Further, the laser irradiation position Lb starts to move from the position Pb2 to the virtual straight line Sv2 simultaneously with the start of the acceleration period Ta, and reaches the virtual straight line Sv2 simultaneously with the end of the acceleration period Ta.

[0180] Further, in a stop period Tt provided on the way during a transition from the acceleration period Ta to the deceleration period Td, both the velocity Vx of the laser irradiation position Lb in the X direction and the velocity Vy thereof in the Y direction become zero and the laser irradiation position Lb is stopped at the position Pb2 with respect to the semiconductor substrate W. In this stop period Tt, the imaging ranges Ri of the imaging parts 8A, 8B are also stopped with respect to the semiconductor substrate W. Particularly, the imaging range Ri of the imaging part 8B is located on the (−X) side of the laser irradiation position Lb located on the (+X) side of the semiconductor substrate W and overlaps the semiconductor substrate W. Accordingly, the infrared camera 81 of the imaging part 8B images a part of the semiconductor substrate W overlapping the imaging range Ri in the stop period Tt (Step S1006).

[0181] Note that, the specific manner of moving the laser irradiation position Lb from the position Pb2 to the virtual straight line Sv2 in the Y direction after moving the laser irradiation position Lb from the virtual straight line Sv1 to the position Pb2 in the Y direction in the switching period Tc is not limited to the example of FIG. 15C, and the movement can be performed in manners shown in FIG. 15D, FIG. 15E and FIG. 15F.

[0182] FIG. 15D shows charts schematically showing a fourth example of the operation performed in accordance with the flow chart of FIG. 14, FIG. 15E shows charts schematically showing a fifth example of the operation performed in accordance with the flow chart of FIG. 14 and FIG. 15F shows charts schematically showing a sixth example of the operation performed in accordance with the flow chart of FIG. 14. Notation in FIGS. 15D to 15F is similar to that in FIG. 15C. FIGS. 15D

to 15F differ from FIG. 15C in a movement mode of the laser irradiation position Lb in the switching period Tc. Accordingly, the following description is centered on differences from FIG. 15C and common operation parts are denoted by corresponding reference signs and description is omitted as appropriate.

[0183] In the fourth example shown in FIG. 15D, the laser irradiation position Lb starts to move in the Y direction from the virtual straight line Sv1 to the position Pb2 simultaneously with the start of the deceleration period Td, and reaches the position Pb2 and stops at the position Pb2 in the Y direction (i.e. the velocity Vy is zero) before the end of the deceleration period Td. However, after the laser irradiation position Lb reaches the position Pb2 in the Y direction, the deceleration period Td continues and the laser irradiation position Lb continues to move in the X direction. Further, the laser irradiation position Lb starts to move in the Y direction from the position Pb2 to the virtual straight line Sv2 after the start of the acceleration period Ta and reaches the virtual straight line Sv2 simultaneously with the end of the acceleration period Ta. That is, in a period ΔT_y from the midway of the deceleration period Td to the midway of the acceleration period Ta, the laser irradiation position Lb is stopped in the Y direction (i.e. the velocity Vy is zero).

[0184] In the fifth example shown in FIG. 15E, the laser irradiation position Lb starts to move in the Y direction from the virtual straight line Sv1 to the position Pb2 simultaneously with the start of the deceleration period Td, and reaches the position Pb2 and stops at the position Pb2 in the Y direction (i.e. the velocity Vy is zero) before the end of the deceleration period Td. However, after the laser irradiation position Lb reaches the position Pb2 in the Y direction, the deceleration period Td continues and the laser irradiation position Lb continues to move in the X direction. Further, the laser irradiation position Lb starts to move in the Y direction from the position Pb2 to the virtual straight line Sv2 simultaneously with the start of the acceleration period Ta, and reaches the virtual straight line Sv2 simultaneously with the end of the acceleration period Ta. That is, in a period ΔT_y from the midway of the deceleration period Td to the start of the acceleration period Ta, the laser irradiation position Lb is stopped in the Y direction (i.e. the velocity Vy is zero).

[0185] In the fifth example shown in FIG. 15F, the laser irradiation position Lb starts to move in the Y direction from the virtual straight line Sv1 to the position Pb2 simultaneously with the start of the deceleration period Td. However, the laser irradiation position Lb has not reached the position Pb2 in the Y direction at the end point of the deceleration period Td. Note that, at the end point of the deceleration period Td, the position (i.e. X-coordinate) of the laser irradiation position Lb and the position (i.e. X-coordinate) of the position Pb2 coincide in the X direction. Accordingly, the laser irradiation position Lb continues to move in the Y direction toward the position Pb2 also after the end of the deceleration period Ta. Further, the laser irradiation position Lb is stopped in the X direction (i.e. the velocity Vx is zero) while the laser irradiation position Lb is moving in the Y direction toward the position Pb2 after the end of the deceleration period Td. The acceleration period Ta is started and the laser irradiation position Lb starts to move in the Y direction from the position Pb2 to the virtual straight line Sv2 at the same time as the laser irradiation position Lb reaches the position Pb2. Further, the laser irradiation position Lb reaches the virtual straight line Sv2 simultaneously with the end of the acceleration period Ta.

[0186] FIG. 15G shows charts schematically showing a seventh example of the operation performed in accordance with the flow chart of FIG. 14. Notation in FIG. 15G is similar to that in FIG. 15A. Also in FIG. 15G, the laser processing is performed in turn for the planned dividing lines S1, S2, S3 in accordance with the flow chart of FIG. 14 as in FIG. 15A. However, FIG. 15G differs from FIG. 15A in the operation in the switching period Tc for changing the planned dividing line S to be laser processed. Accordingly, the following description is centered on differences from FIG. 15A and common operation parts are denoted by corresponding reference signs and description is omitted as appropriate.

[0187] If the laser irradiation position Lb passes through the planned dividing line S1 toward the (+X) side as the laser processing for the planned dividing line S1 is finished, the laser irradiation

position Lb starts to decelerate toward the (+X) side in the X direction (Step S1005) and stops at a position Pb2 on the (+X) side of the semiconductor substrate W in the X direction (Step S1006). This position Pb2 is provided outside a zone between the virtual straight line Sv1 and the virtual straight line Sv2 (on a side opposite to the virtual straight line Sv1 with respect to the virtual straight line Sv2) in the Y direction. That is, in Steps S1005 to S1006, the laser irradiation position Lb moves in the Y direction from the virtual straight line Sv1 to the position Pb2 beyond the virtual straight line Sv2 in parallel with deceleration in the X direction. Further, with the laser irradiation position Lb stopped at the position Pb2, the imaging range Ri of the imaging part 8B stops at a position including at least an imaging point Pw(S3). Accordingly, in Step S1006, the control unit 100 causes the imaging part 8B to image the imaging range Ri and obtains an image including the imaging point Pw(S3). In this way, the control unit 100 can obtain the image showing the position of the unprocessed planned dividing line S3.

[0188] Subsequently, the laser irradiation position Lb stopped at the position Pb2 starts to accelerate toward the (−X) side in the X direction (Step S1001). If the velocity Vx of the laser irradiation position Lb increases to the processing velocity Vxd by the time the laser irradiation position Lb reaches the end of the semiconductor substrate W on the (+X) side, the laser irradiation position Lb moves at the constant processing velocity Vxd toward the (−X) side in the X direction (Step S1002). Further, in a period from the start of acceleration to the start of constant movement at the processing velocity Vxd of the laser irradiation position Lb, the laser irradiation position Lb moves in the Y direction from the position Pb2 to the virtual straight line Sv2. That is, in Steps S1001 to S1002, the laser irradiation position Lb moves in the Y direction from the position Pb2 to the virtual straight line Sv2 in parallel with acceleration in the X direction. In this way, the laser irradiation position Lb reaches the planned dividing line S2 and the line processing for the planned dividing line S2 can be started.

[0189] If the laser irradiation position Lb passes through the planned dividing line S2 toward the (−X) side as the laser processing for the planned dividing line S2 is finished, the laser irradiation position Lb starts to decelerate toward the (−X) side in the X direction (Step S1005) and stops at a position Pb3 on the (−X) side of the semiconductor substrate W in the X direction (Step S1006). This position Pb3 is provided outside a zone between the virtual straight line Sv2 and the virtual straight line Sv3 (on a side opposite to the virtual straight line Sv2 with respect to the virtual straight line Sv3) in the Y direction. That is, in Steps S1005 to S1006, the laser irradiation position Lb moves in the Y direction from the virtual straight line Sv2 to the position Pb2 beyond the virtual straight line Sv3 in parallel with deceleration in the X direction. Further, with the laser irradiation position Lb stopped at the position Pb3, the imaging range Ri of the imaging part 8A stops at a position including at least an imaging point Pw(S4). Accordingly, in Step S1006, the control unit 100 causes the imaging part 8A to image the imaging range Ri and obtains an image including the imaging point Pw(S4). In this way, the control unit 100 can obtain the image showing the position of an unprocessed planned dividing line S4.

[0190] Next, a velocity change of the laser irradiation position Lb is described with reference to “Velocity Change in X Direction” and “Velocity Change in Y Direction” of FIG. 15G. In a line processing period Ts1 (Steps S1002 to S1004) during which the line processing of moving the laser beam B toward the (+X) side along the planned dividing line S1 is performed, the laser irradiation position Lb does not move in the Y direction while moving at the constant processing velocity Vxd in the X direction. Further, in a line processing period Ts2 (Steps S1002 to S1004) during which the line processing of moving the laser beam B toward the (−X) side along the planned dividing line S2 is performed, the laser irradiation position Lb does not move in the Y direction while moving at the constant processing velocity Vxd in the X direction.

[0191] Further, in a switching period Tc (Steps S1005, S1006 and S1001) during which a switch is performed from the line processing period Ts1 to the line processing period Ts2, the laser irradiation position Lb is moved in the Y direction (feeding direction) from the virtual straight line

Sv1 to the virtual straight line Sv2 in parallel with performing the reverse drive in the X direction as in the aforementioned case. Particularly, this movement of the laser irradiation position Lb is made by way of the position Pb2 provided outside the zone between the virtual straight line Sv1 and the virtual straight line Sv2 in the Y direction. That is, the laser irradiation position Lb moves in the Y direction from the virtual straight line Sv1 to the position Pb2 beyond the virtual straight line Sv2 in a deceleration period Td, out of the deceleration period Td and an acceleration period Ta included in the switching period Tc, and moves in the Y direction from the position Pb2 to the virtual straight line Sv2 in the acceleration period Ta. Specifically, the laser irradiation position Lb starts to move from the virtual straight line Sv1 to the position Pb2 simultaneously with the start of the deceleration period Td, and reaches the position Pb2 simultaneously with the end of the deceleration period Td. Further, the laser irradiation position Lb starts to move from the position Pb2 to the virtual straight line Sv2 simultaneously with the start of the acceleration period Ta and reaches the virtual straight line Sv2 simultaneously with the end of the acceleration period Ta.

[0192] Further, in a stop period Tt provided on the way during a transition from the acceleration period Ta to the deceleration period Td, both the velocity Vx of the laser irradiation position Lb in the X direction and the velocity Vy thereof in the Y direction become zero and the laser irradiation position Lb is stopped at the position Pb2 with respect to the semiconductor substrate W. In this stop period Tt, the imaging ranges Ri of the imaging parts **8A**, **8B** are also stopped with respect to the semiconductor substrate W. Particularly, the imaging range Ri of the imaging part **8B** is located on the (-X) side of the laser irradiation position Lb located on the (+X) side of the semiconductor substrate W and overlaps the semiconductor substrate W. Accordingly, the infrared camera **81** of the imaging part **8B** images a part of the semiconductor substrate W overlapping the imaging range Ri in the stop period Tt (Step S1006).

[0193] In the above example, the position Pb2 is provided on the side opposite to the virtual straight line Sv1 with respect to the virtual straight line Sv2 in the Y direction. However, the position Pb2 may be provided on a side opposite to the virtual straight line Sv2 with respect to the virtual straight line Sv1 in the Y direction. In this case, the laser irradiation position Lb moves in the Y direction from the virtual straight line Sv1 to the position Pb2 in the deceleration period Td, and moves in the Y direction from the position Pb2 to the virtual straight line Sv2 beyond the virtual straight line Sv1 in the acceleration period Ta. A similar change can be made also for the position Pb3.

[0194] FIG. **16** is a flow chart showing a first application example of the line processing for each planned dividing line, and FIG. **17** is a chart schematically showing an example of an operation performed in accordance with the flow chart of FIG. **16**. Notation in FIG. **17** is similar to that in FIGS. **15A** to **15G**. The example of FIG. **16** differs from the example of FIG. **14** in the presence of Step S1008 of imaging the semiconductor substrate W during the line processing, but other Steps S1001 to S1007 are common. Accordingly, in the example of FIG. **16**, any one of the operations (first example to seventh example) shown in FIGS. **15A** to **15G** is performed. Note that, although a trace of the laser irradiation position Lb in the switching period Tc is not shown in FIG. **17**, the laser irradiation position Lb can move along the trace shown in any one of FIGS. **15A** to **15G**.

[0195] Step S1008 of FIG. **16** is performed as follows. That is, the semiconductor substrate W is imaged during a movement of the laser irradiation position Lb along the planned dividing line S1 (Step S1008). Specifically, the imaging range Ri (i.e. the imaging range Ri of the imaging part **8A**) located on a moving side (i.e. the (+X) side), toward which the laser irradiation position Lb is moving, relative to the laser irradiation position Lb moving toward the (+X) side is imaged. In this way, an image including an imaging point Pw(S11) on the moving side of the laser irradiation position Lb relative to the laser irradiation position Lb is obtained. In this way, an image showing the position of an unprocessed part, out of the planned dividing line S1 being line processed, can be obtained.

[0196] That is, in a period during which Steps S1003, S1108 and S1104 are performed, the image

of the unprocessed part, out of the planned dividing line **S1** to be line processed, is captured in parallel with performing the line processing for the planned dividing line **S1**.

[0197] Further, the semiconductor substrate **W** is imaged during a movement of the laser irradiation position **Lb** along the planned dividing line **S2** (Step **S1008**). Specifically, the imaging range **Ri** (i.e. the imaging range **Ri** of the imaging part **8B**) located on a moving side (i.e. the $(-X)$ side), toward which the laser irradiation position **Lb** is moving, relative to the laser irradiation position **Lb** moving toward the $(-X)$ side is imaged. In this way, an image including an imaging point **Pw(S21)** on the moving side of the laser irradiation position **Lb** relative to the laser irradiation position **Lb** is obtained. In this way, an image showing the position of an unprocessed part, out of the planned dividing line **S2** being line processed, can be obtained.

[0198] That is, in the period during which Steps **S1003**, **S1108** and **S1104** are performed, the image of the unprocessed part, out of the planned dividing line **S2** to be line processed, is captured in parallel with performing the line processing for the planned dividing line **S2**.

[0199] Furthermore, the semiconductor substrate **W** is imaged during a movement of the laser irradiation position **Lb** along the planned dividing line **S3** (Step **S1008**). Specifically, the imaging range **Ri** (i.e. the imaging range **Ri** of the imaging part **8A**) located on a moving side (i.e. the $(+X)$ side), toward which the laser irradiation position **Lb** is moving, relative to the laser irradiation position **Lb** moving toward the $(+X)$ side is imaged. In this way, an image including an imaging point **Pw(S31)** on the moving side of the laser irradiation position **Lb** relative to the laser irradiation position **Lb** is obtained. In this way, an image showing the position of an unprocessed part, out of the planned dividing line **S3** being line processed, can be obtained.

[0200] That is, in the period during which Steps **S1003**, **S1108** and **S1104** are performed, the image of the unprocessed part, out of the planned dividing line **S3** to be line processed, is captured in parallel with performing the line processing for the planned dividing line **S3**.

[0201] Steps **S1001** to **S1007** are repeated until it is confirmed that the line processing has been completed for the plurality of planned dividing lines **S** (**S1**, **S2**, **S3**, . . .) parallel to the **X** direction (“YES” in Step **S1007**).

[0202] FIG. **18** is a flow chart showing a second application example of the line processing for each planned dividing line, and FIG. **19A** shows charts schematically showing a first example of an operation performed in accordance with the flow chart of FIG. **18**. In FIG. **19A**, a trace of the laser irradiation position **Lb** relatively moving with respect to the semiconductor substrate **W** is shown by dotted line, and the virtual straight lines **Sv1**, **Sv2**, **Sv3** extending in parallel to the **X** direction along the planned dividing lines **S1**, **S2**, **S3** between both outer sides of the planned dividing lines **S1**, **S2**, **S3** are shown by one-dot chain line. Note that dotted lines representing the trace of the laser irradiation position **Lb** are preferentially shown in parts where the trace of the laser irradiation position **Lb** and the virtual straight lines **Sv1**, **Sv2**, **Sv3** overlap.

[0203] In the example shown in FIG. **19A**, the flow chart of FIG. **18** is started from a state where the laser irradiation position **Lb** is stopped at a position **Pb1** on the $(-X)$ side of the semiconductor substrate **W** in the **X** direction. This position **Pb1** is a position provided on the virtual straight line **Sv1** along the planned dividing line **S1**, in other words, facing the planned dividing line **S1** from the **X** direction. However, the position of the laser irradiation position **Lb** when the flow chart of FIG. **18** is started is not limited to this example and can be changed as appropriate.

[0204] In Step **S1101**, the laser irradiation position **Lb** stopped at the position **Pb1** starts to accelerate toward the $(+X)$ side in the **X** direction and moves in parallel to the **X** direction. In this way, the laser irradiation position **Lb** moves toward the $(+X)$ side along the virtual straight line **Sv1**. If the velocity **Vx** of the laser irradiation position **Lb** increases to the processing velocity **Vxd** by the time the laser irradiation position **Lb** reaches the end of the semiconductor substrate **W** on the $(-X)$ side, the laser irradiation position **Lb** moves at the constant processing velocity **Vxd** toward the $(+X)$ side in the **X** direction (Step **S1102**).

[0205] Further, the laser light source **72** is turned on and the irradiation of the laser beam **B** to the

laser irradiation position Lb from the processing head **71** is started in accordance with a timing at which the laser irradiation position Lb reaches the end of the semiconductor substrate W on the (-X) side (Step **S1103**). In this way, the laser beam B is irradiated to the laser irradiation position Lb moving toward the (+X) side in the X direction along the planned dividing line **S1**, whereby the planned dividing line **S1** is processed (line processing).

[0206] Further, in this example, the semiconductor substrate W is imaged during the movement of the laser irradiation position Lb along the planned dividing line **S1** (Step **S1104**). Specifically, the imaging range Ri (i.e. the imaging range Ri of the imaging part **8A**) located on a moving side (i.e. the (+X) side), toward which the laser irradiation position Lb is moving, relative to the laser irradiation position Lb moving toward the (+X) side is imaged. In this way, an image including an imaging point Pw(**S11**) on the moving side of the laser irradiation position Lb relative to the laser irradiation position Lb is obtained. In this way, an image showing the position of an unprocessed part, out of the planned dividing line **S1** being line processed, can be obtained.

[0207] Then, the laser light source **72** is turned off and the irradiation of the laser beam B to the laser irradiation position Lb from the processing head **71** is finished in accordance with a timing at which the laser irradiation position Lb reaches the end of the semiconductor substrate W on the (+X) side (Step **S1105**). In this way, in a period from Step **S1103** to Step **S1105**, the image of the unprocessed part, out of the planned dividing line **S1** to be line processed, is captured in parallel with performing the line processing for the planned dividing line **S1**.

[0208] If the laser irradiation position Lb passes through the planned dividing line **S1** toward the (+X) side, the laser irradiation position Lb starts to decelerate toward the (+X) side in the X direction (Step **S1106**). In Step **S1107**, it is confirmed whether or not the laser processing has been completed for the plurality of planned dividing lines S parallel to the X direction. If there is any unprocessed planned dividing line S, out of these planned dividing lines S (“NO” in Step **S1107**), return is made to Step **S1101**.

[0209] As a result, the laser irradiation position Lb accelerates toward the (-X) side in the X direction (Step **S1101**) after the velocity Vx in the X direction of the laser irradiation position Lb decelerated toward the (+X) side in the X direction becomes zero. If the velocity Vx of the laser irradiation position Lb increases to the processing velocity Vxd by the time the laser irradiation position Lb reaches the end of the semiconductor substrate W on the (+X) side, the laser irradiation position Lb moves at the constant processing velocity Vxd toward the (-X) side in the X direction (Step **S1102**).

[0210] As just described, also in the example of FIGS. **18** and **19A**, the reverse drive is performed in the X direction as in the aforementioned case. Further, in parallel with this reverse drive, the laser irradiation position Lb moves in the Y direction from the virtual straight line Sv1 to the virtual straight line Sv2. In this way, the laser irradiation position Lb reaches the planned dividing line **S2** by moving to the virtual straight line Sv2 in the Y direction by the time the velocity Vx in the X direction of the laser irradiation position Lb increases to the processing velocity Vxd.

[0211] However, in this example, a movement mode in the Y direction of the laser irradiation position Lb is different from the aforementioned one. That is, the laser irradiation position Lb continuously moves in the Y direction from a planned dividing line Sb1 to a planned dividing line Sb2 (continuous feed drive) in parallel with the reverse drive for decelerating, stopping and accelerating the laser irradiation position Lb in the X direction. Particularly, the continuous feed drive of the laser irradiation position Lb in the Y direction is performed throughout the before and after the time at which a timing at which the velocity Vx of the laser irradiation position Lb in the X direction becomes zero due to the reverse drive. Therefore, a timing at which both the velocity Vx in the X direction of the laser irradiation position Lb and the velocity Vy thereof in the Y direction become zero does not exist in this example.

[0212] The laser light source **72** is turned on and the irradiation of the laser beam B to the laser irradiation position Lb from the processing head **71** is started in accordance with a timing at which

the laser irradiation position Lb reaches the end of the semiconductor substrate W on the (+X) side (Step S1103). In this way, the laser beam B is irradiated to the laser irradiation position Lb moving toward the (-X) side in the X direction along the planned dividing line S2 and the planned dividing line S2 is processed (line processing).

[0213] Further, in this example, the semiconductor substrate W is imaged during the movement of the laser irradiation position Lb along the planned dividing line S2 (Step S1104). Specifically, the imaging range Ri (i.e. the imaging range Ri of the imaging part 8B) located on a moving side (i.e. the (-X) side), toward which the laser irradiation position Lb is moving, relative to the laser irradiation position Lb moving toward the (-X) side is imaged. In this way, an image including an imaging point Pw(S21) on the moving side of the laser irradiation position Lb relative to the laser irradiation position Lb is obtained. In this way, an image showing the position of an unprocessed part, out of the planned dividing line S2 being line processed, can be obtained.

[0214] The laser light source 72 is turned off and the irradiation of the laser beam B to the laser irradiation position Lb from the processing head 71 is finished in accordance with a timing at which the laser irradiation position Lb reaches the end of the semiconductor substrate W on the (-X) side (Step S1105). In this way, in a period from Step S1103 to Step S1105, the image of the unprocessed part, out of the planned dividing line S2 to be line processed, is captured in parallel with performing the line processing for the planned dividing line S2.

[0215] If the laser irradiation position Lb passes through the planned dividing line S2 toward the (-X) side, the laser irradiation position Lb starts to decelerate toward the (-X) side in the X direction (Step S1106). In Step S1107, it is confirmed whether or not the line processing has been completed for the plurality of planned dividing lines S parallel to the X direction. If there is any unprocessed planned dividing line S, out of these planned dividing lines S ("NO" in Step S1107), return is made to Step S1101.

[0216] As a result, the laser irradiation position Lb accelerates toward the (+X) side in the X direction (Step S1101) after the velocity Vx in the X direction of the laser irradiation position Lb decelerated toward the (-X) side in the X direction becomes zero. If the velocity Vx of the laser irradiation position Lb increases to the processing velocity Vxd by the time the laser irradiation position Lb reaches the end of the semiconductor substrate W on the (-X) side, the laser irradiation position Lb moves at the constant processing velocity Vxd toward the (+X) side in the X direction (Step S1102).

[0217] At this time, as in the aforementioned case, the continuous feed drive in the Y direction is performed for the laser irradiation position Lb in parallel with the reverse drive in the X direction. In this way, the laser irradiation position Lb reaches the planned dividing line S3 by moving to the virtual straight line Sv3 in the Y direction by the time the velocity Vx in the X direction of the laser irradiation position Lb increases to the processing velocity Vxd.

[0218] The laser light source 72 is turned on and the irradiation of the laser beam B to the laser irradiation position Lb from the processing head 71 is started in accordance with a timing at which the laser irradiation position Lb reaches the end of the semiconductor substrate W on the (-X) side (Step S1103). In this way, the laser beam B is irradiated to the laser irradiation position Lb moving toward the (+X) side in the X direction along the planned dividing line S3 and the planned dividing line S3 is processed (line processing).

[0219] Further, in this example, the semiconductor substrate W is imaged during the movement of the laser irradiation position Lb along the planned dividing line S3 (Step S1104). Specifically, the imaging range Ri (i.e. the imaging range Ri of the imaging part 8A) located on a moving side (i.e. the (+X) side), toward which the laser irradiation position Lb is moving, relative to the laser irradiation position Lb moving toward the (+X) side is imaged. In this way, an image including an imaging point Pw(S31) on the moving side of the laser irradiation position Lb relative to the laser irradiation position Lb is obtained. In this way, an image showing the position of an unprocessed part, out of the planned dividing line S3 being line processed, can be obtained.

[0220] Then, the laser light source **72** is turned off and the irradiation of the laser beam B to the laser irradiation position Lb from the processing head **71** is finished in accordance with a timing at which the laser irradiation position Lb reaches the end of the semiconductor substrate W on the (+X) side (Step **S1105**). In this way, in a period from Step **S1103** to Step **S1105**, the image of the unprocessed part, out of the planned dividing line S3 to be line processed, is captured in parallel with performing the line processing for the planned dividing line S3.

[0221] Next, a velocity change of the laser irradiation position Lb is described with reference to “Velocity Change in X Direction” and “Velocity Change in Y Direction” of FIG. **19A**. In a line processing period Ts1 (Steps **S1103** to **S1105**) during which the line processing of moving the laser beam B toward the (+X) side along the planned dividing line S1 is performed, the laser irradiation position Lb does not move in the Y direction while moving at the constant processing velocity Vxd in the X direction. Further, in a line processing period Ts2 (Steps **S1103** to **S1105**) during which the line processing of moving the laser beam B toward the (−X) side along the planned dividing line S2 is performed, the laser irradiation position Lb does not move in the Y direction while moving at the constant processing velocity Vxd in the X direction.

[0222] Further, in a switching period Tc (Steps **S1106**, **S1101**) during which a switch is performed from the line processing period Ts1 to the line processing period Ts2, the following operation is performed. That is, the X-axis driver **65** (processing-axis driver) performs the reverse drive to bring the laser irradiation position Lb to the planned dividing line S2 (second processing line) by accelerating the laser irradiation position Lb toward the (−X) side (Step **S1101**) after decelerating and stopping the laser irradiation position Lb, which has passed through the planned dividing line S1 (first processing line) toward the (+X) side (first side), toward the (+X) side in the X direction (processing direction) (Step **S1106**). In parallel with this reverse drive, the Y-axis driver **63** (feeding-axis driver) performs the continuous feed drive to continuously move the laser irradiation position Lb in the Y direction (feeding direction) from the virtual straight line Sv1 (first virtual straight line) extended in the X direction to the outside of the planned dividing line S1 along the planned dividing line S1 to the virtual straight line Sv2 (second virtual straight line) extended in the X direction to the outside of the planned dividing line S2 along the planned dividing line S2.

[0223] Particularly, the control unit **100** controls the X-axis driver **65** and the Y-axis driver **63** such that the Y-axis driver **63** starts the continuous feed drive before the X-axis driver **65** stops the laser irradiation position Lb in the X direction by the reverse drive and the Y-axis driver **63** finishes the continuous feed drive after the X-axis driver **65** stops the laser irradiation position Lb in the X direction by the reverse drive. In this way, the Y-axis driver **63** moves the laser irradiation position Lb in the Y direction in a period during which the X-axis driver **65** stops the laser irradiation position Lb in the X direction by the reverse drive.

[0224] In other words, the switching period Tc includes a deceleration period Td (Step **S1006**) for decelerating the laser irradiation position Lb in the X direction and an acceleration period Ta (Step **S1001**) for accelerating the laser irradiation position Lb in the X direction. In contrast, the Y-axis driver **63** continuously performs the movement of the laser irradiation position Lb in the Y direction (i.e. perform the movement without stopping the laser irradiation position Lb in the Y direction) throughout the before and after the time at which a transition period Tx from the deceleration period Td to the acceleration period Ta. Note that the laser irradiation position Lb is stopped in the X direction (i.e. the velocity Vx is zero) during the transition period Tx.

[0225] FIG. **19B** shows charts schematically showing a second example of the operation performed in accordance with the flow chart of FIG. **18**. FIG. **19B** differs from FIG. **19A** in the number of times of imaging the semiconductor substrate W in parallel with the line processing. That is, in the example of FIG. **19B**, the imaging range Ri (i.e. the imaging range Ri of the imaging part **8A**) located on a moving side (i.e. the (+X) side), toward which the laser irradiation position Lb is moving, relative to the laser irradiation position Lb moving toward the (+X) side to perform the line processing for the planned dividing line S1 is imaged a plurality of times (twice in this

example) (Step **S1104**). In this way, two images respectively including two imaging points $Pw(S11)$, $Pw(S12)$ on the moving side of the laser irradiation position Lb relative to the laser irradiation position Lb are obtained. In this way, images showing the positions of unprocessed parts, out of the planned dividing line **S1** being line processed, can be obtained.

[0226] Similarly, the imaging range Ri (i.e. the imaging range Ri of the imaging part **8B**) located on a moving side (i.e. the $(-X)$ side), toward which the laser irradiation position Lb is moving, relative to the laser irradiation position Lb moving toward the $(-X)$ side to perform the line processing for the planned dividing line **S2** is imaged a plurality of times (twice in this example) (Step **S1104**). In this way, two images respectively including two imaging points $Pw(S21)$, $Pw(S22)$ on the moving side of the laser irradiation position Lb relative to the laser irradiation position Lb are obtained. In this way, images showing the positions of unprocessed parts, out of the planned dividing line **S2** being line processed, can be obtained. Further, imaging is similarly performed a plurality of times also in the line processing for the planned dividing line **S3** (Step **S1104**).

[0227] FIG. **20** is a diagram schematically showing an example of an image of the semiconductor substrate obtained in Step **S1008** of FIG. **16** or Step **S1104** of FIG. **18**. In the above examples, a region including an intersection of two planned dividing lines **S** orthogonal to each other is imaged to obtain an image **IM**. At this time, since the image **IM** is obtained while the imaging range Ri is moved in the X direction with respect to the semiconductor substrate **W**, a luminance is shown to be averaged in the X direction in the image **IM**. As a result, a high-luminance region having a high luminance and extending in parallel to the X direction in correspondence with the planned dividing line **S** and low-luminance regions having a luminance lower than the high-luminance region and extending in parallel to the X direction in correspondence with the semiconductor chips **C** appear. Particularly, in the Y direction, the high-luminance region is sandwiched by two low-luminance regions. Therefore, the control unit **100** can confirm the position in the Y direction of the planned dividing line **S** based on the high-luminance region corresponding to the planned dividing line **S**.

[0228] FIG. **21** is a flow chart showing an example of a determination method of laser processing conditions in the line processing, FIG. **22A** show charts showing parameters relating to the determination of laser processing conditions, FIG. **22B** is a graph showing a time impact of the laser processing condition, and FIG. **22C** is a table showing an example of a table to be referred to in the determination of the laser processing conditions of FIG. **21**. This table is stored in the storage **190** in advance.

[0229] FIG. **22A** show an upper chart showing a relationship of the velocity Vx of the laser irradiation position Lb moving in the X direction and time and a lower chart showing a relationship of the velocity Vx of the laser irradiation position Lb moving in the X direction and the position in the X direction (i.e. X -coordinate) of the laser irradiation position Lb in a line processing.

[0230] As shown in the lower chart, to perform the line processing for the planned dividing line **S**, an irradiation position scan is performed to irradiate the laser beam **B** to the laser irradiation position Lb overlapping the planned dividing line **S** while moving the laser irradiation position Lb in the X direction from a start point Xs on one side of the planned dividing line **S** to an end point Xe on the other side (side opposite to the one side) of the planned dividing line **S**. That is, in the irradiation position scan, the laser beam **B** is irradiated from the processing head **71** to the laser irradiation position Lb overlapping the planned dividing line **S** while the laser irradiation position Lb is moved in the X direction from the start point Xs to the end point Xe by the X -axis driver **65**. In this way, the aforementioned line processing is performed in accordance with the irradiation position scan.

[0231] In this irradiation position scan, a constant velocity zone **SC** is set for the planned dividing line **S**. This constant velocity zone **SC** is located between the start point Xs and the end point Xe in the X direction and set to include the planned dividing line **S**. In this example, both ends of the constant velocity zone **SC** coincide with both ends of the planned dividing line **S** in the X direction,

in other words, the constant velocity zone SC coincides with the planned dividing line S. However, a setting mode of the constant velocity zone SC is not limited to this example and the constant velocity zone SC may be set by adding offsets outside the both ends of the planned dividing line S. In this case, the constant velocity zone SC becomes longer than the planned dividing line S. A length of the offset may be a predetermined fixed value or may be a value obtained by multiplying a length of the planned dividing line S by a predetermined scale factor (e.g. 1%). The length of such a constant velocity zone SC is set according to that of the planned dividing line S.

Specifically, the longer the planned dividing line S, the longer the constant velocity zone SC (in other words, the shorter the planned dividing line S, the shorter the constant velocity zone SC). [0232] In this irradiation position scan, the laser irradiation position Lb moves in the X direction from the start point Xs provided on one side of the constant velocity zone SC to the end point X3 provided on the other side of the constant velocity zone SC. Further, in the acceleration period Ta during which the laser irradiation position Lb moves in the X direction from the start point Xs to an end Xss on the one side of constant velocity zone SC in the X direction, the laser irradiation position Lb accelerates at an acceleration A in the X direction and the velocity Vx in the X direction of the laser irradiation position Lb increases from zero to the processing velocity Vxd. Further, in a constant velocity period Tsc (coinciding with the line processing period Ts in this example) during which the laser irradiation position Lb moves in the X direction from the end Xss on the one side of the constant velocity zone SC to an end Xse on the other side of the constant velocity zone SC, the laser irradiation position Lb moves at the constant processing velocity Vxd in the X direction. Further, in the deceleration period Tb during which the laser irradiation position Lb moves in the X direction from the end Xse on the other side of the constant velocity zone SC to the end point Xe, the laser irradiation position Lb decelerates at the acceleration A in the X direction and the velocity Vx in the X direction of the laser irradiation position Lb decreases from the processing velocity Vxd to zero.

[0233] At this time, the acceleration period Ta is a period (V_{xd}/A) required to increase the velocity Vx from zero to the processing velocity Vxd at the acceleration A, the constant velocity period Tsc is a period (L_{sc}/V_{xd}) required to move a constant velocity distance Lsc, which is the length of the constant velocity zone SC, at the processing velocity Vxd, and the deceleration period Td is a period (V_{xd}/A) required to decrease the velocity Vx from the processing velocity Vxd to the zero at the acceleration A. Therefore, a scanning time t required for the irradiation position scan is:

$$t=2\times V_{xd}/A+L_{sc}/V_{xd}.$$

[0234] Thus, a relationship shown in FIG. 22B is satisfied between the processing velocity Vxd and the scanning time t. That is, the scanning time t has a minimum value when the processing velocity Vxd is $V_{xd_min} (= (L_{sc} \times A / 2).sup.1/2)$. Therefore, the line processing can be efficiently performed by setting the processing velocity Vxd according to the length (constant velocity distance Lsc) of the constant velocity zone SC.

[0235] However, in the case of changing the processing velocity Vxd, a frequency of the laser beam B emitted from the laser light source 72 needs to be changed. Specifically, as the processing velocity Vxd is increased, the frequency of the laser beam B needs to be increased. In contrast, the frequency of the laser beam B can be changed only stepwise and cannot be continuously changed. Accordingly, the table of FIG. 22C is used. This table specifies a relationship of the constant velocity distance Lsc (length of the planned dividing line S in this example), the processing velocity Vxd and the frequency fc of the laser beam B. Specifically, the following laser processing conditions are set in the table. If the constant velocity distance Lsc is less than or equal to Lsc(1), the processing velocity Vxd is set to Vxd(1) and the frequency of the laser beam B is set to fc(1). If the constant velocity distance Lsc is more than Lsc(1) and less than or equal to Lsc(2), the processing velocity Vxd is set to Vxd(2) and the frequency of the laser beam B is set to fc(2).

[0236] That is, in the laser processing condition determination of FIG. 21, the length (constant

velocity distance L_{sc}) of the constant velocity zone SC set for the planned dividing line S to be line processed is obtained (Step S1201). Then, the processing velocity V_{xd} is determined (Step S1202) and the frequency f_c of the laser beam B is determined (Step S1203) based on the constant velocity distance L_{sc} obtained in Step S1201 and the table of FIG. 22C. The irradiation position scan is performed according to the laser processing conditions (processing velocity V_{xd} and frequency f_c) determined by FIG. 21 in this way.

[0237] The irradiation position scan is performed in turn for the plurality of planned dividing lines S parallel to the X direction. In other words, a plurality of the irradiation position scans are performed for the planned dividing lines S different from each other. In contrast, the laser processing condition determination of FIG. 21 is performed for each of the plurality of irradiation position scans. In each irradiation position scan, the laser irradiation position L_b is moved and the laser beam B is irradiated according to the laser processing conditions determined for each irradiation position scan.

[0238] Particularly, if the semiconductor substrate W formed with the plurality of planned dividing lines S parallel to the X direction is circular as in the above example, the further from a center of a circle in the Y direction, the shorter the planned dividing line S becomes, and shorter the constant velocity distance L_{sc} set for the planned dividing line S also becomes. That is, the constant velocity distance L_{sc} set for the irradiation position scan differs depending on the position in the Y direction of the planned dividing line S, for which the irradiation position scan is performed. Accordingly, it is proper to perform the laser processing condition determination for each irradiation position scan performed in turn for the plurality of planned dividing lines S.

[0239] Note that the laser processing condition determination can be performed at an arbitrary timing before the start of the irradiation position scan which is a target of the laser processing condition determination. For example, the laser processing condition determination may be performed for all of the plurality of irradiation position scans before the plurality of irradiation position scans respectively corresponding to the plurality of planned dividing lines S parallel to the X direction are started. Alternatively, in the case of performing the next irradiation position scan following one irradiation position scan, the laser processing condition determination may be performed for the next irradiation position scan during the execution of the one irradiation position scan.

[0240] In the embodiment described above, the planned dividing line S is processed by irradiating the laser beam B from the processing head 71 to the laser irradiation position L_b moving along the targeted planned dividing line S (one target line), out of the plurality of planned dividing lines S (processing lines), while moving the laser irradiation position L_b with respect to the semiconductor substrate W from the start point X_s on one side to the end point X_e on the other side of the semiconductor substrate W (processing object) in the X direction (processing direction) (irradiation position scan). In this irradiation position scan, the constant velocity zone SC including the planned dividing line S (target line) in the X direction is set between the start point X_s and the end point X_e . In the acceleration period T_a (first period) during which the laser irradiation position L_b moves from the start point X_s to the end X_{ss} of the constant velocity zone SC on the one side, the velocity of the laser irradiation position L_b in the X direction with respect to the semiconductor substrate W increases from zero to the processing velocity V_{xd} . Further, the laser irradiation position L_b moves at the constant processing velocity V_{xd} in the X direction with respect to the semiconductor substrate W through the constant velocity period T_{sc} (second period) during which the laser irradiation position L_b moves from the end X_{ss} on the one side to the end X_{se} on the other side of the constant velocity zone SC. Furthermore, in the deceleration period T_b (third period) from the arrival of the laser irradiation position L_b at the end X_{se} of the constant velocity zone SC on the other side to the arrival thereof at the end point X_e , the velocity of the laser irradiation position L_b in the X direction with respect to the semiconductor substrate W decreases from the processing velocity V_{xd} to zero. At this time, the length (constant velocity distance L_{sc}) of the constant

velocity zone SC is set according to the length of the planned dividing line S in the X direction, and the processing velocity Vxd in the irradiation position scan is adjusted according to the length (constant velocity distance Lsc) of the constant velocity zone SC (velocity adjustment processing, Step S1202). Therefore, it is possible to suppress a total time (scanning time t) of the acceleration period Ta, the constant velocity period Tsc and the deceleration period Td. In this way, the planned dividing line S can be efficiently processed in the laser processing technique for processing the planned dividing line S by moving the laser beam B along the planned dividing line S of the semiconductor substrate W.

[0241] Further, the control unit **100** performs the plurality of irradiation position scans by repeating the irradiation position scan while changing the targeted planned dividing line S (target line) among the plurality of planned dividing lines S. Then, the control unit **100** performs the velocity adjustment processing (Step S1202) for each of the plurality of irradiation position scans. In such a configuration, each of the plurality of planned dividing lines S can be efficiently processed, and the processing of the semiconductor substrate W can be quickly completed.

[0242] Further, the control unit **100** performs a frequency adjustment processing (Step S1203) of adjusting a frequency of the laser beam B to be irradiated to the laser irradiation position Lb moving along the targeted planned dividing line S (target line) in the irradiation position scan according to the processing velocity Vxd in the irradiation position scan. In such a configuration, the planned dividing line S can be precisely processed by irradiating the laser beam B of a suitable frequency corresponding to the adjusted processing velocity Vxd to the planned dividing line S.

[0243] Further, the control unit **100** performs a plurality of the irradiation position scans by repeating the irradiation position scan while changing the targeted planned dividing line S (target line) among the plurality of planned dividing lines S. Then, the control unit **100** performs the frequency adjustment processing (Step S1203) for each of the plurality of irradiation position scans. In such a configuration, each of the plurality of planned dividing lines S can be precisely processed.

[0244] Note that, as shown in FIG. 22C, the processing velocity Vxd is adjusted by selecting one, out of a plurality of discrete processing velocities Vxd(1), Vxd(2), Vxd(3) and Vxd(4), and the oscillation frequency fc is adjusted by selecting one, out of a plurality of discrete oscillation frequencies fc(1), fc(2), fc(3) and fc(4). That is, in the laser processing condition determination, the processing velocity Vxd and the oscillation frequency fc are selected based on to which of a plurality of (four) ranges shown in FIG. 22C the constant velocity distance Lsc belongs. At this time, if the ranges, to which the constant velocity distance Lsc belongs, are the same between two laser irradiation position scans successively performed when the processing velocity Vxd and the oscillation frequency fc are adjusted by performing the laser processing condition determination for each of the plurality of irradiation position scans, the processing velocity Vxd and the oscillation frequency fc are maintained. On the other hand, if the ranges, to which the constant velocity distance Lsc belongs, are different between two laser irradiation position scans successively performed, the processing velocity Vxd and the oscillation frequency fc are changed (in other words, switched). That is, the adjustment of the processing velocity Vxd includes the maintaining of the processing velocity Vxd and the change (switch) of the processing velocity Vxd, and the adjustment of the oscillation frequency fc includes the maintaining of the oscillation frequency fc and the change (switch) of the oscillation frequency fc.

[0245] In the above example, the laser processing condition determination of FIG. 21 is performed for each irradiation position scan performed for one planned dividing line S. However, the laser processing conditions (the processing velocity Vxd and the frequency fc of the laser beam B) need not be determined for one irradiation position scan, but may be determined for one semiconductor substrate W. That is, the control unit **100** may perform the following flow chart of FIG. 23, instead of the flow chart of FIG. 21.

[0246] FIG. 23 is a flow chart showing an example of the laser processing condition determination

method for each semiconductor substrate, and FIG. 24 is a diagram showing an example of a table to be referred to in the laser processing condition determination of FIG. 23. This table is stored in the storage 190 in advance. In this example, a count value n for counting the plurality of planned dividing lines S parallel to the X direction and included in the semiconductor substrate W held on the chuck stage 3 is added. For example, the count value n takes a value of 1 to 20 if the semiconductor substrate W has twenty planned dividing lines S parallel to the X direction. [0247] The count value n is reset to zero in Step S1301, and incremented in Step S1302. In Step S1303, a constant velocity distance Lsc(n), which is a length of the constant velocity zone SC set for the n.sup.th planned dividing line S, is obtained. Then, a scanning time t(n) required for the irradiation position scan for the n.sup.th planned dividing line S is calculated based on the following equation (Step S1304):

$$t(n)=2 \times Vxd/A+Lsc(n)/Vxd.$$

[0248] In Step S1305, it is judged whether or not the count value n has reached a maximum value nx (equivalent to the number of the planned dividing lines S parallel to the X direction and included in the semiconductor substrate W) thereof. Unless the count value n has reached the maximum value nx (“NO” in Step S1305), Steps S1302 to S1304 are repeated. By repeating Steps S1302 to S1304 in this way until the count value n reaches the maximum value nx, the scanning times t(n) are calculated for all the plurality of planned dividing lines S parallel to the X direction and included in the semiconductor substrate W.

[0249] If the count value N reaches the maximum value nx (“YES” in Step S1305), a total scanning time Tt1 is calculated based on the following equation (Step S1306):

$$Tt1=t(1)+t(2)+ \dots t(nx-1)+t(nx)=\Sigma t(n).$$

[0250] This total scanning time Tt1 is a function of the processing velocity Vxd. In contrast, the processing velocity Vxd that results in a minimum total scanning time Tt1 is calculated as an optimal processing velocity Vxg in step S1307. In Step S1308, the processing conditions (the processing velocity Vxd and the frequency fc of the laser beam B) are determined using the thus calculated optimal processing velocity Vxg and the table of FIG. 24.

[0251] The table of FIG. 24 specifies the optimal processing velocity Vxg, the processing velocity Vxd and the frequency fc of the laser beam B. Specifically, the following laser processing conditions are specified in the table. If the optimal processing velocity Vxg is equal to or less than Vxg(1), the processing velocity Vxd is set to Vxd(11) and the frequency of the laser beam B is set to fc(11). If the optimal processing velocity Vxg is more than Vxg(1) and equal to or less than Vxg(2), the processing velocity Vxd is set to Vxd(12) and the frequency of the laser beam B is set to fc(12). Similarly to the table of FIG. 22C, the faster the processing velocity Vxd, the higher the frequency of the laser beam B according to the table of FIG. 24.

[0252] That is, in this example, the processing velocity Vxd and the frequency fc of the laser beam B are not changed through an execution period of the irradiation position scans for the plurality of planned dividing lines S parallel to the X direction and included in one semiconductor substrate W held on the chuck stage. In this way, it is possible to eliminate an influence of a time required to change the frequency fc of the laser beam B on the processing completion of the semiconductor substrate W.

[0253] Note that the optimal processing velocity Vxd is correlated to the constant velocity distance Lsc set for each of the nx planned dividing lines S. The above control of setting the processing velocity Vxd based on such an optimal processing velocity Vxd is equivalent to a processing of adjusting the processing velocity Vxd in the irradiation position scan according to the length (constant velocity distance Lsc) of the constant velocity zone SC set for each of the nx planned dividing lines S. Further, the above control of determining the frequency fc of the laser beam B based on such an optimal processing velocity Vxd is equivalent to a process of adjusting the

frequency f_c of the laser beam B to be irradiated to the laser irradiation position Lb according to the processing velocity V_{xd} in the irradiation position scan.

[0254] As just described, the control unit **100** sets the processing velocity V_{xd} (common processing velocity) common to the plurality of irradiation position scans performed by repeating the irradiation position scan while changing the planned dividing line S among the plurality of planned dividing lines S (Step **S1308**). Each of the plurality of irradiation position scans is performed based on the processing velocity V_{xd} set in Step **S1308**. Particularly, in Step **S1308** (velocity adjustment processing), the processing velocity V_{xd} is adjusted according to the total scanning time T_{t1} reflecting the constant velocity distance $L_{sc}(n)$, which is the length of the constant velocity zone SC in each of the plurality of irradiation position scans. In such a configuration, the plurality of planned dividing lines S can be efficiently processed without switching the processing velocity V_{xd} among the plurality of planned dividing lines S, and the processing of the semiconductor substrate W can be quickly completed.

[0255] Further, the control unit **100** calculates the frequency f_c common to the plurality of irradiation position scans according to the processing velocity V_{xd} commonly set for the plurality of planned dividing lines S (Step **S1308**). In each of the plurality of irradiation position scans, the laser beam B of the common frequency f_c is irradiated to the laser irradiation position Lb. In such a configuration, the frequency f_c of the laser beam B needs not be switched among the plurality of planned dividing lines S, and it is possible to eliminate an influence of a time required to switch the frequency f_c on the processing completion of the semiconductor substrate W.

[0256] Note that, as shown in FIG. **24**, the processing velocity V_{xd} is adjusted by selecting one, out of a plurality of discrete processing velocities $V_{xd}(1)$, $V_{xd}(2)$, $V_{xd}(3)$ and $V_{xd}(4)$, and the oscillation frequency f_c is adjusted by selecting one, out of a plurality of discrete oscillation frequencies $f_c(1)$, $f_c(2)$, $f_c(3)$ and $f_c(4)$. That is, in the laser processing condition determination, the processing velocity V_{xd} and the oscillation frequency f_c are selected based on to which of a plurality of (four) ranges shown in FIG. **24** the optimal processing velocity V_{xd} belongs. At this time, if the ranges, to which the optimal processing velocity V_{xd} belongs, are the same between two semiconductor substrates W to be successively processed when the processing velocity V_{xd} and the oscillation frequency f_c are adjusted by performing the laser processing condition determination for each of a plurality of semiconductor substrate W, the processing velocity V_{xd} and the oscillation frequency f_c are maintained. On the other hand, if the ranges, to which the optimal processing velocity V_{xd} belongs, are different between two semiconductor substrates W to be successively processed, the processing velocity V_{xd} and the oscillation frequency f_c are changed (in other words, switched). That is, the adjustment of the processing velocity V_{xd} includes the maintaining of the processing velocity V_{xd} and the change (switch) of the processing velocity V_{xd} , and the adjustment of the oscillation frequency f_c includes the maintaining of the oscillation frequency f_c and the change (switch) of the oscillation frequency f_c .

[0257] In the example described above, the processing velocity V_{xd} and the frequency f_c of the laser beam B are not changed through the execution period of the irradiation position scans for the plurality of planned dividing lines S parallel to the X direction and included in one semiconductor substrate W held on the chuck stage **3**. In other words, the laser processing conditions (the processing velocity V_{xd} and the frequency f_c of the laser beam B) are adjusted not on per-planned dividing line S basis, but on per-semiconductor substrate W basis. At this time, diameters of the semiconductor substrates W used in the laser processing apparatus **1** may be supposed. In such a case, the processing conditions (the processing velocity V_{xd} and the frequency f_c of the laser beam B) corresponding to each diameter may be stored in the storage **190** by performing the laser processing condition determination method of FIG. **23** for the diameters of these semiconductor substrates W in advance.

[0258] FIG. **25** is a flow chart showing an example of laser processing condition setting to set the processing conditions obtained in advance by the laser processing condition method of FIG. **23**

according to the diameter of the substrate, and FIG. 26 is a diagram schematically showing a correspondence relationship table used in the flow chart of FIG. 25. In this example, the processing velocity V_{xd} and the frequency f_c of the laser beam B are obtained by performing the laser processing condition determination of FIG. 23 in advance for each of the semiconductor substrate W having a diameter of 200 mm and the semiconductor substrate W having a diameter of 300 mm used in the laser processing apparatus 1. The processing velocity V_{xd} and the frequency f_c of the laser beam B obtained for the diameter of each semiconductor substrate W are stored as a correspondence relationship table in the storage 190.

[0259] That is, in the correspondence relationship table Tsvf of FIG. 26, the processing velocity V_{xd} ($=V_{200}$) and the frequency f_c ($=f_{c_200}$) of the laser beam B are associated with the semiconductor substrate W having the diameter of 200 mm, and the processing velocity V_{xd} ($=V_{300}$) and the frequency f_c ($=f_{c_300}$) of the laser beam B are associated with the semiconductor substrate W having the diameter of 300 mm.

[0260] In Step S1401, the laser processing control calculator 120 obtains the size (diameter) of the semiconductor substrate W as a target of the irradiation position scan. Then, the laser processing control calculator 120 reads out the processing velocity V_{xd} corresponding to the size of the semiconductor substrate W obtained in Step S1401 from the correspondence relationship table Tsvf, and sets the read-out processing velocity V_{xd} as the processing velocity V_{xd} during the execution of the irradiation position scan (Step S1402). Further, the laser processing control calculator 120 reads out the frequency f_c of the laser beam B corresponding to the size of the semiconductor substrate W obtained in Step S1401 from the correspondence relationship table Tsvf, and sets the read-out frequency f_c as the frequency f_c during the execution of the irradiation position scan (Step S1403).

[0261] In such a configuration, for example, in the case of performing the irradiation position scan for the semiconductor substrate W of 200 mm, the irradiation position scan for each of the plurality of planned dividing lines S parallel to the X direction and included in the semiconductor substrate W held on the chuck stage 3 is performed under such laser processing conditions that the processing velocity V_{xd} is the processing velocity V_{200} and the frequency f_c of the laser beam B is f_{c_200} .

[0262] In the above example, the correspondence relationship table Tsvf (correspondence relationship information) representing a correspondence relationship of the diameter, the processing velocity V_{xd} (common processing velocity) and the frequency f_c (common frequency) for each of the plurality of diameters (sizes) of the semiconductor substrates W is stored in the storage 190. Then, the laser processing control calculator 120 (control unit) adjusts the processing velocity V_{xd} and the frequency f_c of the laser beam B based on the diameter of the semiconductor substrate W as a target of the irradiation position scan and the correspondence relationship table Tsvf (Steps S1402, S1403). In such a configuration, the laser processing control calculator 120 can easily adjust the processing velocity V_{xd} and the frequency f_c of the laser beam B by referring to the correspondence relationship table Tsvf.

[0263] As just described, in the above embodiment, the laser processing apparatus 1 corresponds to an example of a “laser processing apparatus” of the disclosure, the chuck stage 3 corresponds to an example of a “supporting member” of the disclosure, the X-axis driver 65 corresponds to an example of a “processing-axis driver” of the disclosure, the processing head 71 corresponds to an example of a “processing head” of the disclosure, the laser light source 72 corresponds to an example of a “laser light source” of the disclosure, the control unit 100 corresponds to an example of a “control unit” of the disclosure, the control unit 100 corresponds to an example of a “computer” of the disclosure, the laser processing program 191 corresponds to an example of a “laser processing program” of the disclosure, the recording medium 192 corresponds to an example of a “recording medium” of the disclosure, the laser beam B corresponds to an example of a “laser beam” of the disclosure, the laser irradiation position Lb corresponds to an example of a “laser

irradiation position” of the disclosure, the planned dividing line S corresponds to an example of a “processing line” of the disclosure, the constant velocity zone SC corresponds to an example of a “constant velocity zone” of the disclosure, the acceleration period Ta corresponds to an example of a “first period” of the disclosure, the deceleration period Td corresponds to an example of a “third period” of the disclosure, the constant velocity period Tsc corresponds to an example of a “second period” of the disclosure, the processing velocity Vxd corresponds to an example of a “processing velocity” of the disclosure, the semiconductor substrate W corresponds to an example of a “processing object” of the disclosure, the X direction corresponds to an example of a “processing direction” of the disclosure, the end point Xe corresponds to an example of an “end point” of the disclosure, the start point Xs corresponds to an example of a “start point” of the disclosure, the end Xss corresponds to an example of an “end on one side” of the disclosure, the end Xse corresponds to an example of an “end on the other side” of the disclosure, Step S1202 corresponds to an example of a “velocity adjustment processing” of the disclosure, and Step S1203 corresponds to an example of a “frequency adjustment processing” of the disclosure.

[0264] Note that the disclosure is not limited to the above embodiment and various changes other than those described above can be made without departing from the gist of the disclosure. For example, the use of the captured image is not particularly described in the above embodiment. However, such an image can be used in various uses. For example, the unprocessed planned dividing line S may be displaced in the Y direction according to the laser processing for the planned dividing line S. Accordingly, the control unit **100** can calculate a displacement amount of the unprocessed planned dividing line S in the Y direction based on the captured image of the semiconductor substrate W and can align the planned dividing line S to be line processed and the laser irradiation position Lb based on this displacement amount.

[0265] Further, although the imaging part **8** images the intersection of the two planned dividing lines S orthogonal to each other in the above examples, an object to be imaged by the imaging part **8** is not limited to this and may be, for example, an alignment mark or the like attached to the semiconductor chip C.

[0266] Further, the specific configuration for relatively moving the laser irradiation position Lb with respect to the semiconductor substrate W is not limited to the XYθ drive table **6** and may be, for example, a driving mechanism for driving the processing head **71** in the X direction and the Y direction.

[0267] Further, the number of the imaging parts **8** is not limited to two and may be, for example, one.

[0268] Further, the individually separated semiconductor chips C may be manufactured by the laser processing method (substrate processing of FIG. **11** or the like) described above (semiconductor chip manufacturing method). In this semiconductor chip manufacturing method, a modified layer is formed by performing the line processing for the planned dividing lines S of the semiconductor substrate W by the above laser processing method (laser processing step). Subsequently, each of the plurality of semiconductor chips C is separated by stretching and expanding the tape E holding the semiconductor substrate W (expanding step).

Claims

1. A laser processing apparatus, comprising: a supporting member configured to support a processing object having a plurality of processing lines parallel to each other such that the processing lines are parallel to a predetermined processing direction; a processing head including a laser light source configured to emit a laser beam, the processing head being configured to irradiate the laser beam to a laser irradiation position from the laser light source; a processing-axis driver configured to relatively move the laser irradiation position in the processing direction with respect to the processing object by driving at least one of the supporting member and the processing head

in the processing direction; and a controller configured to perform an irradiation position scan to process one target line, out of the plurality of processing lines, by irradiating the laser beam from the processing head to the laser irradiation position moving along the one target line while causing the processing-axis driver to move the laser irradiation position with respect to the processing object from a start point on one side of the processing object to an end point on the other side opposite to the one side of the processing object in the processing direction, in the irradiation position scan, a constant velocity zone including the target line being set between the start point and the end point in the processing direction, a velocity of the laser irradiation position in the processing direction with respect to the processing object increasing from zero to a processing velocity in a first period of moving the laser irradiation position from the start point to an end of the constant velocity zone on the one side, the laser irradiation position moving at the constant processing velocity in the processing direction with respect to the processing object through a second period of moving the laser irradiation position from the end on the one side to an end on the other side of the constant velocity zone, and the velocity of the laser irradiation position in the processing direction with respect to the processing object decreasing from the processing velocity to the zero in a third period from arrival of the laser irradiation position at the end of the constant velocity zone on the other side to arrival of the laser irradiation position at the end point, and the controller being configured to set a length of the constant velocity zone according to a length of the target line in the processing direction and performing a velocity adjustment processing of adjusting the processing velocity in the irradiation position scan according to the length of the constant velocity zone.

2. The laser processing apparatus according to claim 1, wherein: the controller is configured to perform a plurality of the irradiation position scans by repeating the irradiation position scan while changing the target line among the plurality of processing lines and perform the velocity adjustment processing for each of the plurality of irradiation position scans.

3. The laser processing apparatus according to claim 1, wherein: the controller is configured to perform a frequency adjustment processing of adjusting a frequency of the laser beam to be irradiated to the laser irradiation position moving along the target line in the irradiation position scan according to the processing velocity in the irradiation position scan.

4. The laser processing apparatus according to claim 3, wherein: the controller is configured to perform a plurality of the irradiation position scans by repeating the irradiation position scan while changing the target line among the plurality of processing lines and perform the frequency adjustment processing for each of the plurality of irradiation position scans.

5. The laser processing apparatus according to claim 1, wherein: the controller is configured to set a common processing velocity, the common processing velocity being a processing velocity common to a plurality of the irradiation position scans performed by repeating the irradiation position scan while changing the target line among the plurality of processing lines, and perform each of the plurality of irradiation position scans with the processing velocity set to the common processing velocity, and the common processing velocity is adjusted according to the length of the constant velocity zone in each of the plurality of irradiation position scans in the velocity adjustment processing.

6. The laser processing apparatus according to claim 5, wherein: the controller is configured to obtain a common frequency common to the plurality of irradiation position scans according to the common processing velocity and cause the laser beam of the common frequency to be irradiated to the laser irradiation position in each of the plurality of irradiation position scans.

7. The laser processing apparatus according to claim 6, further comprising: a storage configured to store correspondence relationship information representing a correspondence relationship of a size, the common processing velocity and the common frequency for each of a plurality of sizes of the processing objects, wherein: the controller is configured to adjust the common processing velocity and the common frequency based on the size of the processing object as a target of the irradiation

position scan and the correspondence relationship information.

8. The laser processing apparatus according to claim 7, wherein: the correspondence relationship information represents a diameter of the processing object as the size and represents the correspondence relationship of the size, the common processing velocity and the common frequency for each of a diameter of 200 mm and a diameter of 300 mm.

9. The laser processing apparatus according to claim 1, wherein: both ends of the constant velocity zone coincide with both ends of the target line in the processing direction.

10. A laser processing method, comprising: supporting a processing object having a plurality of processing lines parallel to each other by a supporting member such that the processing lines are parallel to a predetermined processing direction; and performing an irradiation position scan to process one target line, out of the plurality of processing lines, by irradiating a laser beam to a laser irradiation position from a processing head including a laser light source for emitting the laser beam, the processing head irradiating the laser beam from the laser light source to the laser irradiation position, while moving the laser irradiation position with respect to the processing object from a start point on one side of the processing object to an end point on the other side, being opposite to the one side, of the processing object, in the processing direction by a processing-axis driver for relatively moving the laser irradiation position in the processing direction with respect to the processing object by driving at least one of the processing head and the supporting member in the processing direction, in the irradiation position scan, a constant velocity zone including the target line being set between the start point and the end point in the processing direction, a velocity of the laser irradiation position in the processing direction with respect to the processing object increasing from zero to a processing velocity in a first period of moving the laser irradiation position from the start point to an end of the constant velocity zone on the one side, the laser irradiation position moving at the constant processing velocity in the processing direction with respect to the processing object through a second period of moving the laser irradiation position from the end on the one side to an end on the other side of the constant velocity zone, and the velocity of the laser irradiation position in the processing direction with respect to the processing object decreasing from the processing velocity to the zero in a third period from arrival of the laser irradiation position at the end of the constant velocity zone on the other side to arrival of the laser irradiation position at the end point, a length of the constant velocity zone being set according to a length of the target line in the processing direction, and the processing velocity in the irradiation position scan being adjusted according to the length of the constant velocity zone.

11. The laser processing method according to claim 10, further comprising: setting a common processing velocity, the common processing velocity being a processing velocity common to a plurality of the irradiation position scans performed by repeating the irradiation position scan while changing the target line among the plurality of processing lines, wherein: each of the plurality of irradiation position scans is performed with the processing velocity set to the common processing velocity, and the common processing velocity is adjusted according to the length of the constant velocity zone in each of the plurality of irradiation position scans.

12. A non-transitory computer readable medium storing a laser processing program for causing a computer to carry out the laser processing method according to claim 10.

13. (canceled)

14. A semiconductor chip manufacturing method, comprising: processing a semiconductor substrate, having a plurality of semiconductor chips demarcated by processing lines and arrayed, by the laser processing method according to claim 10; and separating each of the plurality of semiconductor chips by expanding a tape, holding the semiconductor substrate by an adhesive force, processed by the laser processing method.

15. A semiconductor chip, manufactured by: processing a semiconductor substrate, having a plurality of semiconductor chips demarcated by processing lines and arrayed, by the laser processing method according to claim 10; and separating each of the plurality of semiconductor

- chips by expanding a tape, holding the semiconductor substrate by an adhesive force, processed by the laser processing method.
- 16.** The laser processing apparatus according to claim 2, wherein: the controller is configured to perform a frequency adjustment processing of adjusting a frequency of the laser beam to be irradiated to the laser irradiation position moving along the target line in the irradiation position scan according to the processing velocity in the irradiation position scan.
- 17.** The laser processing apparatus according to claim 2, wherein: both ends of the constant velocity zone coincide with both ends of the target line in the processing direction.
- 18.** The laser processing apparatus according to claim 3, wherein: both ends of the constant velocity zone coincide with both ends of the target line in the processing direction.
- 19.** A non-transitory computer readable medium storing a laser processing program for causing a computer to carry out the laser processing method according to claim 11.
- 20.** A semiconductor chip manufacturing method, comprising: processing a semiconductor substrate, having a plurality of semiconductor chips demarcated by processing lines and arrayed, by the laser processing method according to claim 11; and separating each of the plurality of semiconductor chips by expanding a tape, holding the semiconductor substrate by an adhesive force, processed by the laser processing method.
- 21.** A semiconductor chip, manufactured by: processing a semiconductor substrate, having a plurality of semiconductor chips demarcated by processing lines and arrayed, by the laser processing method according to claim 11; and separating each of the plurality of semiconductor chips by expanding a tape, holding the semiconductor substrate by an adhesive force, processed by the laser processing method.
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